Per core, adjoint gradients reach validated torque-ripple targets 2.5 and 4.1 times sooner than finite differences in a nonlinear motor model

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Abstract

Field gradients pay off in design only if they shorten the path to a design that survives independent validation. We reconstruct a published nonlinear permanent-magnet motor in our GEODE finite-element implementation and compare five optimizers in a preregistered, budget-matched study, in an eight- and an 18-parameter shape space, judged by one common order-93 validation model, whose interface order exceeds the training model's because truncation lowers loaded torque ripple by 7–9% and raises cogging by 59%. SLSQP with adjoint gradients reached torque-ripple targets sooner than the derivative-free COBYQA at the same training resolution in all 45 paired comparisons where both succeeded, by geometric means of 2.45–3.25 and 3.14–7.79 in the two spaces. A post-hoc ablation, not preregistered but run under a protocol frozen beforehand, kept SLSQP and replaced the adjoint by forward differences whose stencil solves start Newton-cold. With every run on one single-threaded core, the adjoint reached the primary target 2.47 and 4.11 times sooner, from every start, and 5.58 and 10.07 times sooner once the shared validation charges are excluded. The remaining factor, COBYQA against finite-difference SLSQP, is 1.2 in the eight-parameter space, ranging from 0.70 to 2.27 across starts, and 1.9 in the 18-parameter space. A second post-hoc arm with warm-started stencils tests the strongest finite-difference competitor: [PENDING exp2c-fd-warm] Every selected target crossing met its target on a finer mesh. The adjoint, more than the gradient-based optimizer, accounts for SLSQP's advantage.

1 Introduction

Whether to pay for derivatives is a practical design decision. An adjoint gives the sensitivities of a few outputs to many parameters for a cost that does not grow with the number of parameters, whereas a finite-difference gradient needs one extra solve per parameter (Giles and Pierce, 2000; Martins and Hwang, 2013). That does not settle whether following adjoint gradients reaches a good design sooner than a derivative-free search or a cheaper gradient, or whether the design survives a better model: the adjoint differentiates the chosen discrete model and excitation policy. In aerodynamic shape optimization an adjoint gradient-based method has needed far fewer function evaluations than a genetic algorithm (Zingg et al., 2008), so the direction of an advantage is expected. Its size for a nonlinear motor, measured on a validated target with failures retained, is the question here.

We answer it in two steps. A preregistered, budget-matched study compares five methods in two nested design spaces of a published nonlinear permanent-magnet motor (Wiesheu et al., 2024), judges every candidate on one validation model finer than any training model, and charges whole-run wall time with nothing subtracted (Section 4). There SLSQP with adjoint gradients beats COBYQA, but the contrast changes the search strategy and the gradient together. A post-hoc

ablation therefore keeps SLSQP and replaces the adjoint by forward differences (Section 5). With the stencil solved serially and Newton-cold, on one core per run, the adjoint reaches the target 2.47 and 4.11 times sooner, a ratio set mainly by the cost of the gradient, since both variants cross each target after the same number of validated designs. A second post-hoc arm with warm-started stencils tests the strongest finite-difference competitor: [PENDING exp2c-fd-warm]. What is left of the main contrast, COBYQA against finite-difference SLSQP, is small and start-dependent with eight parameters and about a factor of two with 18: in the smaller space, gradient-based search without an adjoint is not reliably faster than the derivative-free method.

Two checks make these measurements interpretable. The ripple objective depends on the rotor–stator interface model: truncating harmonic coupling at the reference's order 45 changes loaded ripple by 7–9% and cogging by 59% on the finest mesh, so the study validates at order 93 (Section 3). The design gradients pass finite-difference and Taylor checks, yet verified ripple sensitivities differ between interface/estimator models, and the excitation rule reverses mean-torque sensitivities (Section 6). The numerical methods are established; the contribution is the measured time-to-target evidence for adjoint gradients, separated from the choice of optimizer, with the checks that show what the gradients represent. Section 7 states what transfers.

2 Related work

Wiesheu et al. (2024) combine spline-based rotor and stator shape optimization with engineering parameters for a nonlinear permanent-magnet synchronous motor, and their related study combines parameter and shape optimization with isogeometric analysis (Wiesheu et al., 2025). Their public implementation and saved outputs provide the reproduction target. We inherit the initial geometry, material table and excitation conventions and replace the field discretization; the comparison is between numerical models, not with a measured motor.

Adjoint shape sensitivity for nonlinear magnetostatics is long established. Park et al. (1992) applied the adjoint variable method to the finite-element equations of a saturated two-dimensional magnet design. Gangl et al. (2015) derive rotor shape sensitivities with a shape-Lagrangian formulation to smooth the rotation pattern of a motor; Kuci et al. (2017) derive shape sensitivities from the Lie derivative and apply them to an interior permanent-magnet machine (Kuci et al., 2016). Scotzniovsky et al. (2025) combine free-form deformation, differentiable mesh warping and adjoint derivatives, and Merkel et al. (2021) use isogeometric shape derivatives with harmonic rotor–stator coupling. Martins and Hwang (2013) review finite differences, complex steps, algorithmic differentiation and direct and adjoint methods in one framework, Giles and Pierce (2000) introduce the adjoint approach to design, Nikolova et al. (2004) review adjoint-variable sensitivities for electromagnetic structure design, and Peter and Dwight (2010) and Kenway et al. (2019) survey sensitivity-analysis approaches and adjoint practice for aerodynamic optimization. These works establish why an adjoint gradient is cheaper than a finite-difference one; Section 5 measures what that saving is worth in time to a validated target. The accuracy of a finite-difference gradient also depends on its step relative to the function's computational noise, which Moré and Wild (2011) show how to estimate and Moré and Wild (2012) use to choose a near-optimal step; our finite-difference arm instead uses one step fixed in advance.

Interface order already matters for torque: De Gersem and Weiland (2004) formulate harmonic weighting functions at the sliding interface, Bontinck et al. (2018) use harmonic interface constraints between isogeometric subdomains and analyze their stability, and Egger et al. (2022) derive torque from a harmonic mortar formulation and vary the transmitted order in a linear cogging experiment. We quantify its effect on loaded nonlinear ripple and design sensitivities, using coupling torque and the gap-averaged stress of Arkkio (1987); Howe and Zhu (1992) and Arkkio and Hannukainen (2012)

study discretization effects on torque.

Optimizer benchmarks in machine design compare derivative-free methods: Duan and Ionel (2013) compare response-surface and differential-evolution search on a five-variable permanent-magnet motor, and Bramerdorfer et al. (2018) survey optimization practice for electrical machines, and Lei et al. (2017) review their design-optimization methods. Ion et al. (2018) compare deterministic gradient-based and stochastic search for electric devices, and Zingg et al. (2008) found that a genetic algorithm needed 5 to 200 times as many function evaluations as an adjoint method in aerodynamic shape optimization. On an eight-variable wing twist problem, Yu et al. (2018) found gradient-free optimizers 4 to over 100 times as expensive as adjoint-driven ones at the same optimum, the counterpart of our eight-parameter SLSQP-against-COBYQA ratios of 3.10–3.25. They do not compare adjoint with finite-difference gradients, which our ablation does, but state its premise: finite-difference cost grows with the number of variables and adjoint cost does not. Our time-to-target measure follows the performance profiles of Dolan and Moré (2002) and the data profiles of Moré and Wild (2009), with a wall-time budget and censored targets.

3 Motor model

3.1 Geometry and operating states

The full machine has six poles and 36 stator slots. We model a 60° sector, occupying $60° \le \theta \le 120°$ at zero rotation, with antiperiodic conditions on the radial boundaries. The rotor inner and outer radii are 16 and 44 mm; the stator inner and outer radii are 45 and 67.5 mm. Axial length is $L = 0.1$ m. The 1 mm air gap is split at the coupling radius $r_c = 44.5$ mm. Figure 1 shows the material-tagged reconstruction.

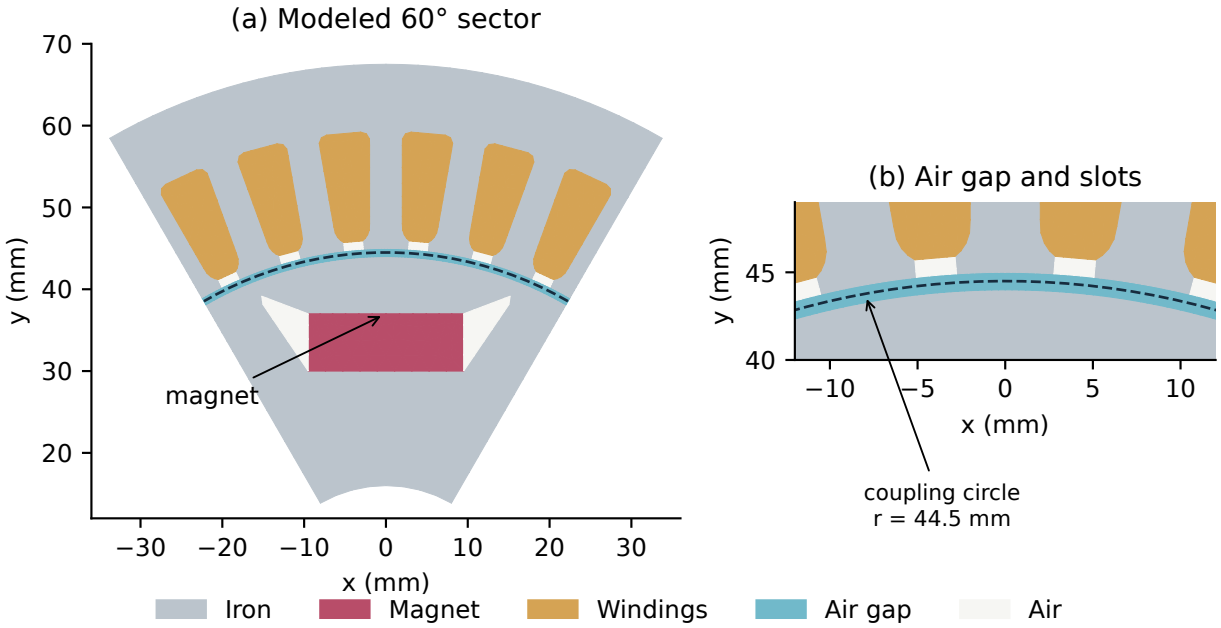


Figure 1: Initial-design geometry rendered directly from the committed coarsest mesh. The dashed line is the rotor–stator coupling circle. The right panel enlarges the gap and slot openings. Material boundaries are shown without the element edges; the geometry is not an optimized result.

The reconstruction uses the initial NURBS exports in the public snapshot. Their control points are stored with weights already multiplied in, and patch orientation must be accounted for when

integrating areas. Two thin rotor patches described as air slits in source comments receive iron under the reference's material assignment; we follow that assignment. The mesh generator, built on Gmsh 4.15.2 (Geuzaine and Remacle, 2009), preserves the winding groups separately, since their two characteristic areas are approximately 36.29 and 31.22 mm^2.

At mechanical angle β, phase currents are

$$I_q(\beta) = aI_* \sin(\phi_0 + 3\beta + \delta_q), \qquad (\delta_U, \delta_V, \delta_W) = (0, -2\pi/3, 2\pi/3), \tag{1}$$

where $a \in \{0, 0.5, 1\}$, $I_* = 9.677657374\,\mathrm{A}$, and the electrical phase offset $\phi_0 = 0$ for the initial design. Angles in Eq. (1) are in radians. Each winding group carries current density $J_g = 12 s_g I_q / |\Omega_g|$, with its own area and prescribed sign $s_g \in \{-1, 1\}$. Using a common area would change the current density between groups by about 16%. We compare 20 distinct mechanical angles, $0°, 1°, \ldots, 19°$, excluding the duplicated endpoint in the source CSV. The reference's optimization grid contains the ten even angles; both grids are retained in the artifacts.

3.2 Field model and rotor–stator coupling

We solve two-dimensional nonlinear magnetostatics for the vector potential A_z with continuous piecewise-linear triangles and a tabulated M27 iron law, using Newton's method with the consistent material tangent. The rotor and stator meet at the coupling circle in one of two ways: a conforming interface identifies matching trace nodes after rotation, while the harmonic model, following the reference, imposes trace continuity only for the orders $n = 3, 9, \ldots, N_h$ through Lagrange multipliers, with $N_h = 45$ unless stated otherwise. Torque is evaluated either by the gap-averaged Arkkio expression (Arkkio, 1987) or, for the harmonic model, by the coupling torque from the constrained energy derivative (Egger et al., 2022). Forward comparisons use Arkkio for both couplings to isolate the interface change; the derivative study pairs harmonic coupling with coupling torque and conforming coupling with Arkkio; the optimization study trains on order-45 coupling torque and validates on order 93. Retaining every order below the trace Nyquist limit reproduces the conforming Arkkio torques within $4.1 \times 10^{-9}\,\mathrm{N\,m}$ on L0 and $1.3 \times 10^{-9}\,\mathrm{N\,m}$ on L1, without fitting any parameter to the saved torque. Appendix A gives the weak form, tangent, iron law, coupling constraints and torque expressions.

3.3 Meshes and statistics

The study uses four meshes, L0–L3, with 3,153, 9,937, 36,083 and 139,268 nodes. The 60° coupling arc has $120 \cdot 2^{\ell}$ segments at level ℓ, so integer-degree rotation is represented at every level. A solution at one angle initializes the next angle at the same current. We report the mean $\bar{T}$ and the population standard deviation σ_T of the torque over the $N = 20$ angles; σ_T is the ripple metric. At zero current it measures cogging, the torque variation caused by the permanent magnets and slots without applied phase current. The squared-ripple objective is $F = \sum_{a \in \{0, 0.5, 1\}} \sigma_T(a)^2$, in $\mathrm{N^2\,m^2}$, on the 20-angle grid. Appendix A lists the implementation checks.

3.4 Interface order sets the validation model

Table 1 compares the same geometry, material and excitation with two interface models. The conforming L3 means are 0.15% and 0.16% low at half and full current. Harmonic coupling reduces the mean deviations to −0.002% and +0.051%, with both reference means inside the estimated refinement intervals. The loaded ripple deviations fall from +9.9% and +11.3% to +1.7% and +1.3%.

Spatial refinement at fixed order and order refinement on a fixed mesh answer different questions (Fig. 2; Appendix E defines the refinement interval). At L3, order 45 raises cogging standard

Table 1: L3 population standard deviation and mean torque, in N m per sector, on the 20-angle grid. The last column gives ripple deviation from the saved reference; n/a identifies the reference itself. The reference is subsequently refined in Section E.1.

Current fraction	Model	Mean	Std	Std vs ref. (%)
0	Reference	0.00010	0.01071	n/a
0	Conforming	0.00001	0.00760	−29.1
0	Harmonic	0.00000	0.01209	+12.9
0.5	Reference	0.66208	0.03895	n/a
0.5	Conforming	0.66106	0.04279	+9.9
0.5	Harmonic	0.66206	0.03961	+1.7
1	Reference	1.44248	0.07935	n/a
1	Conforming	1.44023	0.08829	+11.3
1	Harmonic	1.44322	0.08036	+1.3

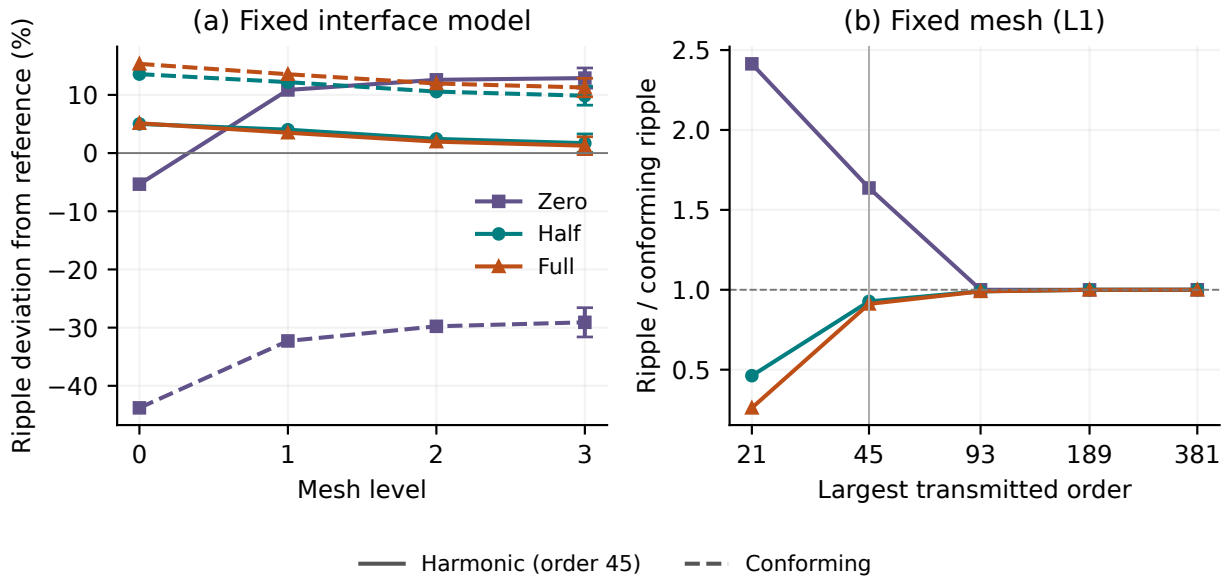


Figure 2: Left: ripple deviation from the reference under spatial refinement, with solid lines for harmonic order 45 and dashed lines for conforming coupling; L3 bars use Eq. (11). Right: order refinement on L1 with logarithmic order spacing, normalized by the conforming ripple at that same level. The saved reference uses order 45; see Section E.1 for its refinement. Colors identify current, not coupling.

deviation by about 59% and lowers loaded standard deviation by 7–9%, relative to the conforming values. At L1 and L2, order 189 approaches the conforming torque to within 1.1×10^{-4} N m. A spatially refined order-45 solution thus remains a solution of a truncated interface model. Matching that truncation improves reproduction of the published calculation; it does not establish which model better represents the physical motor.

Rerunning the public reference with its own isogeometric model at four refinement levels reduces its cogging discrepancy with GEODE from 12.9% to 0.10%, so the coarse saved reference explains the initial cogging gap at matched order, although neither refinement sequence is fully converged. For the saved optimized design of the reference, raising the order from 45 to 93 multiplies its squared-ripple objective by 3.6. Appendix E gives these checks, the estimator and material controls, and the saved design's size and current normalizations.

4 Matched optimization study

This section tests whether adjoint-driven optimization shortens the time needed to reach a common validated ripple target. The protocol and all of its amendments were fixed in the preregistration before any main-run outcome was observed. One later deviation, in how the final refinement was run, was adopted after the main runs had finished and is described with the refinement in Section 4.2; it changes no selection rule, check or target.

4.1 Protocol

Methods. Five methods form an optimizer-by-resolution comparison plus one surrogate (Table 2). A and E are SLSQP (Kraft, 1988) using the field adjoints of Section 6, safeguarded so that a refused request restores the last admissible point and restarts the solver; B and D are COBYQA (Ragonneau, 2022) with $2n+1$ interpolation points and unit-box trust radii 0.1 to 10^{-6}, where n is the number of free parameters. Both run through SciPy 1.18.1 (Virtanen et al., 2020). C is an empirical trust-region method on four cubic radial-basis interpolants of the L1 training outputs (the three per-current standard deviations and the full-current mean), refitted after every evaluation from an $n+1$-point initial design; Appendix B gives its radii, acceptance and update rules and the other settings. A/B and E/D vary the optimizer at fixed training resolution, and A/E and B/D vary resolution at fixed optimizer. The optimizer contrast changes the search strategy and the gradient source together; Section 5 separates them. Training uses order-45 coupling torque on 20 angles at current fractions 0, $\frac{1}{2}$ and 1 of the fixed peak current 9.6777 A. L0 and L1 differ only in spatial resolution, and neither is fitted or corrected against validation. Time to a fixed target, with unattained targets kept as censored, follows the data- and performance-profile practice for benchmarking optimization solvers (Dolan and Moré, 2002; Moré and Wild, 2009). The preregistration also named a fixed group-1 current-density rule as a labelled control. It was not run in this study, which uses the fixed physical current throughout; Section 6 reports that rule's effect on sensitivities.

Table 2: Methods. D and E solve the coarse nonlinear field equations; they are neither fitted magnetic circuits nor reduced-basis models.

Method	Optimizer or search model	Training mesh
A	Safeguarded SLSQP, field adjoints	L1
B	COBYQA, forward evaluations	L1
C	Empirical cubic-RBF trust region	L1
D	COBYQA, forward evaluations	L0
E	Safeguarded SLSQP, field adjoints	L0

Design spaces and constraints. Space A has eight free geometric lengths: MAG, MH, MW, SW1–SW4 and Sr1 (Appendix J), that is, the rotor iron depth, magnet height and width, four stator tooth and slot dimensions, and the slot-bottom fillet radius. Space B adds ten grouped offsets acting on 55 surface control points. Stack length, radial scale, stator outer diameter and operating angle are fixed, so the saved Opt120 design lies outside both spaces. The two spaces also use different discrete mesh-motion maps, so their contrast does not isolate the ten added offsets. Each training model $r \in \{\mathrm{L0}, \mathrm{L1}\}$ minimizes $F_r = \sum_c \sigma_{r,c}^2$ over the three training currents, subject to $\mu_{r,1} \geq 1.0025\,\mu_{r,1}(\theta_0)$ for its own full-current mean and to normalized construction margins $c_i - 0.05 \geq 0$. Unbuildable or mesh-infeasible candidates are refused: SLSQP restores toward the last admissible point and restarts, COBYQA receives a NaN barrier, and C skips the point. A moved mesh with an inverted triangle or a minimum angle below 5° is repaired by up to four charged

remeshing attempts. Because refinement cannot remove a limiting geometric corner, the 5° policy also acts as an implicit geometric constraint. Space B's slot-opening channel wall is not guarded and can become very thin at extreme offsets; that risk is handled by repair and refusal, not by a wall-thickness constraint.

The gradients A and E use, including those of space B's grouped offsets, agree with central differences in both spaces within 2.8×10^{-7} relatively in the directional checks (Appendix C).

Common validation and credit. Every method is judged on the same order-93, L2, 20-angle validation model. Its initial-design anchor is $F_0 = 9.6034 \times 10^{-3}\,\mathrm{N^2\,m^2}$ with full-current mean $\mu_0 = 1.44084\,\mathrm{N\,m}$. The targets are $T_{50} = 0.5F_0$, $T_{90} = 0.1F_0$ and $T_{99} = 0.01F_0$. A validated design attains a target only if

$$F_{\mathrm{val}} + \delta F \le T, \qquad \delta F = \max\{0.10F_{\mathrm{val}},\, 2.5 \times 10^{-7}\,\mathrm{N^2\,m^2}\}, \tag{2}$$

its validated full-current mean is at least $1.0025\,\mu_0$, its construction margins and mesh checks pass, and its validation completes within budget. The allowance δF is an empirical margin, not a bound on continuum error. Optimizer success flags confer no credit. Changed incumbents are validated as they appear, after the initial interpolation set for B, C and D.

Budget and timing. Cost is wall time, in units of the median cold validation forward measured on the same host at the same concurrency. Each run receives 120 units, with a 60-unit checkpoint that counts only validations completed by then. The charge covers evaluator start-up, geometry checks, mesh motion and repair, training solves, adjoints, optimizer overhead, surrogate fitting and validation; nothing is subtracted as overhead. L0 receives no discount.

Starts and execution. Each space uses θ_0 and four Latin-hypercube starts (seeds 1–4), with the scalar coordinates matched across spaces. A start was eligible only if it was buildable and admissible, and its moved reference mesh met the mesh policy without repair, in both spaces at L0 and L1. The starts therefore favor moderate deformations and are not uniform samples of the design box. All starts were frozen before any main outcome. Every method runs from every start twice, giving 100 runs. Within a start, repeat 1 rotates the method order and repeat 2 reverses it.

Each run is one single-threaded process; 15 ran concurrently on one AWS c7a.4xlarge host, and all 100 completed in one session without recovery or host contamination (Appendix C).

4.2 Results

Repeats. The two repeats of each run follow the same optimization path. In all 50 pairs, the sequences of validated objectives agree exactly up to the shorter run's length, and they differ in length only when the budget or a solver stop cuts one repeat earlier. The repeats therefore measure timing noise, not independent outcomes, and the comparison is a paired case study over five starts. Over the 111 target crossings attained in both repeats, the time differs by a median of 2.1% and at most 12.4% relative to repeat 1. Repeat-1 times are reported below.

Attained quality. Figure 3 shows the best credited value of $(F + \delta F)/F_0$ for every start, method and space (repeat 1). In space A, no run goes below $F/F_0 = 0.043$ and none reaches T_{99}. The best designs of A, B, D and E all reach $0.043F_0$, and C's reaches $0.049F_0$; in Fig. 3, which includes the allowance, the A, B, D and E level appears near 0.048 (C's near 0.054). All five lie within 0.17% above the full-current mean floor, so the mean constraint is active there: the level shared by A, B, D and E is the constrained optimum they found in the eight-parameter space, not a property of any

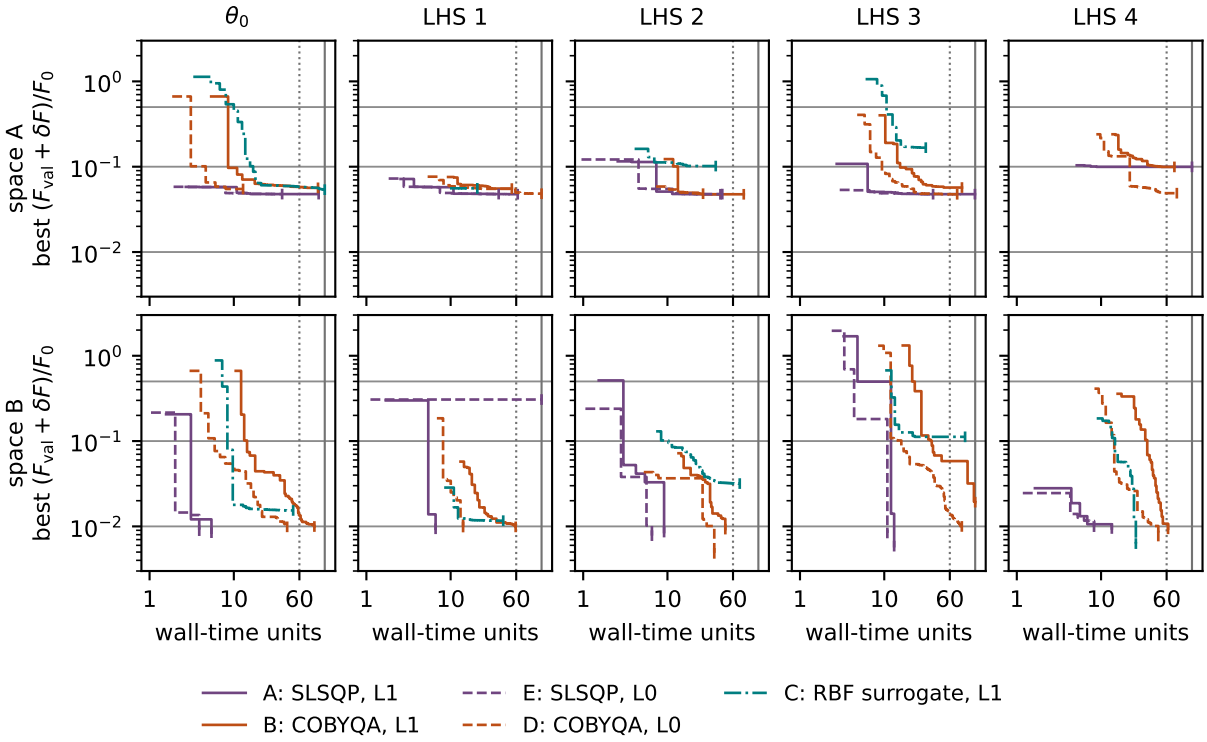


Figure 3: Best credited $(F_{\text{val}} + \delta F)/F_0$ versus charged wall time, repeat 1, at common L2/order-93/20-angle validation. Grey lines mark T_{50}, T_{90} and T_{99}; the dotted and solid vertical lines mark the 60-unit checkpoint and the 120-unit budget. The time axis is logarithmic. Each curve ends at the run's actual stop (tick). A missing curve means the run had no credited validation.

one method. In space A the start mattered more than the method. From LHS 4, A and B stop near $0.09F_0$, D reaches $0.044F_0$, and C and E never produce a credited validation, in either repeat. In space B, each method's best design reaches 0.004–$0.009F_0$, and every method except C reaches T_{99} from at least four of five starts.

Time to target. Figure 4 gives the charged time to each target, per start, and Table 4 in Appendix D lists the values. In repeat 1, only three crossings occur after the 60-unit checkpoint, all at T_{99} in space B: two by B and one by D.

Table 5 (Appendix D) gives paired time ratios. A ratio is formed only when both methods attain the target from that start; no finite ratio is derived from a censored cell. Wherever both reach a target from the same start, gradient-based SLSQP gets there sooner than COBYQA at the same training resolution. At L1 (B/A), the geometric-mean ratio is 3.10–3.25 in space A and 5.06–7.79 in space B. At L0 (D/E), it is 2.45–2.75 and 3.14–7.20. All 45 finite optimizer-contrast pairs (by space, target and start) favor SLSQP, with ratios from 1.46 to 21.5. The coarse mesh shortens the time by a smaller factor: the geometric-mean E/A ratio is 0.62–0.79 and D/B is 0.45–0.56. As charged, A is faster than D, the inexpensive derivative-free physical competitor, by a geometric mean of 1.68–1.83 in space A and 2.25–4.32 in space B. This contrast changes both factors; one of its 25 pairs is a near tie (0.99, space-A LHS 2 at T_{90}: D is 0.75% faster, within the timing noise between repeats), and in space A most of the contrast comes from validation charges (next paragraph). We conjecture that part of the larger space-B ratios comes from COBYQA's initial interpolation set, which costs

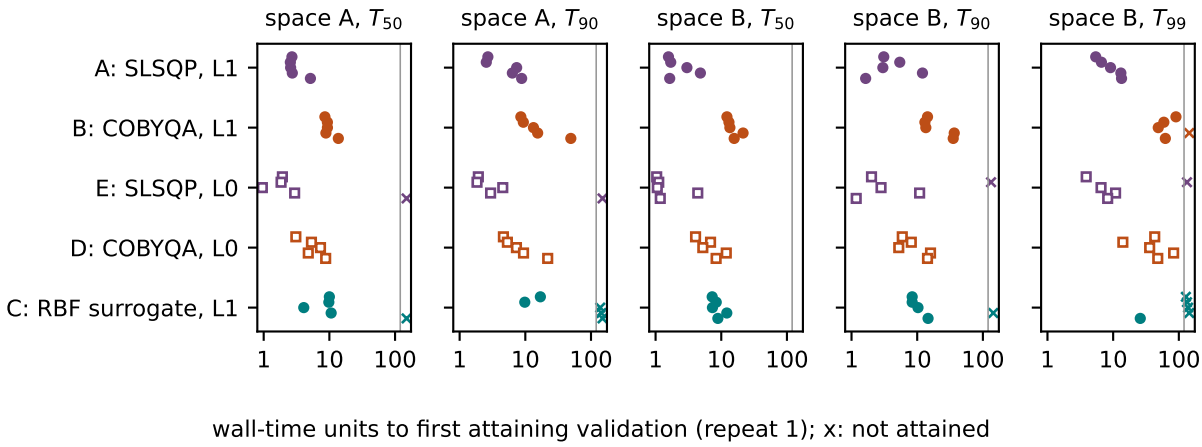


Figure 4: Charged wall time to the first attaining validation, repeat 1, one marker per start, for every target attained by some run. Filled markers: L1 training; open markers: L0 training; crosses: not attained within 120 units.

$2n + 1 = 37$ training evaluations before its first model step there, against 17 in space A.

Validation charges. Validating every new incumbent makes validation the largest single charge (Table 7 in Appendix D). One validation costs a median of 0.87 units over the 3,350 completed validations (range 0.55–1.10), and validation takes 53% of all charged time, 2–90% per run. SLSQP crosses a target after few validations, but most of its time to a target is validation; COBYQA at L1 (B) needs more validations, which are a smaller part of its time. Excluding the validation charges up to each crossing, the same-resolution geometric means rise to 5.41–5.66 (B/A) and 5.55–5.56 (D/E) in space A, and to 8.84–13.58 and 7.92–15.27 in space B, so the ratios as charged are conservative for the optimizer contrast. The coarse mesh cuts the remaining time more, by geometric means of 0.19–0.27 (E/A) and 0.20–0.23 (D/B), but it does not make validation cheaper, which is why its overall saving is only 1.3–2.2 times. The space-A contrast between A and D does not survive this decomposition: excluding validation, its geometric mean is 1.15–1.16, and two of its ten space-A pairs favor D (0.97 and 0.54). In space A, A's lead over D therefore reflects mainly how many validated incumbents each method needs before crossing under the validate-every-incumbent rule (medians of two and six). In space B the non-validation D/A mean remains 1.92–3.19.

The speed of E has a reliability cost. E never produces a credited validation from space-A LHS 4. From space-B LHS 1 it reaches T_{50} but then stalls at $0.28F_0$, missing T_{90} and T_{99}. In four start–target cells, D reaches a target that E misses. A, by contrast, reaches every target that any method reaches, from every start. C attains T_{99} only once in space B and reaches T_{90} from two of five space-A starts.

Final refinement. The preregistered selection takes, for each method and space, the lowest-F_{val} feasible design over both repeats, and the earliest repeat-1 crossing of each attained target. With θ_0 in each space and identical designs merged, this gives 33 designs. Each received the six registered checks: the primary L2 validation, L3, a 40-angle grid, Newton tolerance 10^{-12}, an independently solved grid shifted by 0.5°, and held-out currents 0.25 and 0.75. The 15 designs with zero stator offsets also received the conforming outer-band Arkkio check on fresh meshes. All 213 checks completed (Table 6 in Appendix D). These checks qualify the selected designs only; they do not check each of the study's 111 repeat-1 target crossings.

The refinement used a hash-verified copy of the study outputs on a different host (Appendix C). Spatial refinement is the largest change. Over all 33 designs, L3 moves F by -4.0% to $+6.3\%$. Over the 25 target crossings alone, the range is -4.0% to $+1.1\%$. The largest increase belongs to the overall best design, D's in space B, whose F rises from $0.0044F_0$ to $0.0047F_0$. Angular and nonlinear effects are small: the 40-angle and shifted grids change F by at most 0.11%, and the tighter Newton tolerance by less than 1.1×10^{-5} relative. The conforming estimator gives 1.1% to 4.4% more ripple than the order-93 coupling on the space-A designs, and 2.0–2.1% more at θ_0. All 25 target crossings still satisfy their target, allowance included, at L3; the smallest remaining margin is 1.2% of the target. Every candidate also satisfies the full-current mean floor in the primary, L3, 40-angle, tight and shifted checks, the closest by 0.012%. The refinement therefore confirms the selected crossings and leaves the time-to-target comparison unchanged. At held-out currents 0.25 and 0.75, part of space B's extra reduction is lost (Appendix D).

5 Post-hoc ablation: adjoint against finite-difference gradients

The contrast of Section 4 changes the search strategy and the gradient source together. This ablation changes only how SLSQP obtains its gradient, and therefore what the gradient costs. It was added after the main outcomes were known and is not part of the preregistered study; its public protocol was frozen before any of its runs (Code and data availability).

Protocol. Method F is the safeguarded SLSQP of A with identical bounds, constraints, restoration, validation policy, targets, mean floor and stopping rules. Only the gradients of the objective and of the mean constraint change: they are forward differences with a step of 10^{-6} in each coordinate's physical unit (metres for the lengths and offsets), reusing the base value, so one gradient costs n extra training solves ($n = 8$ in space A and 18 in space B). Each stencil solve is an ordinary training solve on the base point's mesh generation, and like every training solve in the preregistered training configuration it starts Newton-cold (all 40 traces record a cold start), so it costs as much as any training solve. A component whose forward step is unusable (outside the box, unbuildable or on another mesh generation) takes the backward step, as 8% of components did; no gradient was refused. Every solve is charged, and the geometry-constraint Jacobians remain analytic. Every run, F and A′ alike, is one single-threaded process, so F's n stencil solves run one after another and the time ratio is also a ratio of compute; solving the stencil in parallel would shorten F's wall time per gradient but not its compute. A is rerun unchanged in the same session as A′, so the comparison shares host, time unit and load: both methods run from the five starts, in both spaces, with two repeats and rotated order, giving 40 runs of 120 units on the L1 training model. The primary estimand is, per start, the repeat-1 ratio F/A′ of time to T_{90} in space A and to T_{99} in space B; a censored cell gives no finite ratio.

Results. Both F and A′ reach T_{50} and T_{90} in space A and all three targets in space B from all five starts, in both repeats; neither reaches T_{99} in space A, and no cell is censored. The adjoint reaches the primary target sooner from every start (Fig. 5): the geometric-mean F/A′ ratio is 2.47 in space A (range 2.45–2.52) and 4.11 in space B (3.84–4.33), and repeat 2 gives 2.49 and 4.04 (Table 8). Both cross each target after the same number of validations from every start. Up to the primary crossing their validated objectives agree within 0.4% in space A and within 0.9% at three space-B starts; at space-B LHS 2 and LHS 4 the iterates drift apart before the T_{99} crossing, by 29% at the crossing validation and 18% one validation before it, and at LHS 2 F uses one training evaluation fewer, yet each crossing comes at the same validation. The gap is therefore mainly the cost of the gradient. Costs are in the session's unit, the median cold validation solve (66.66 s). One

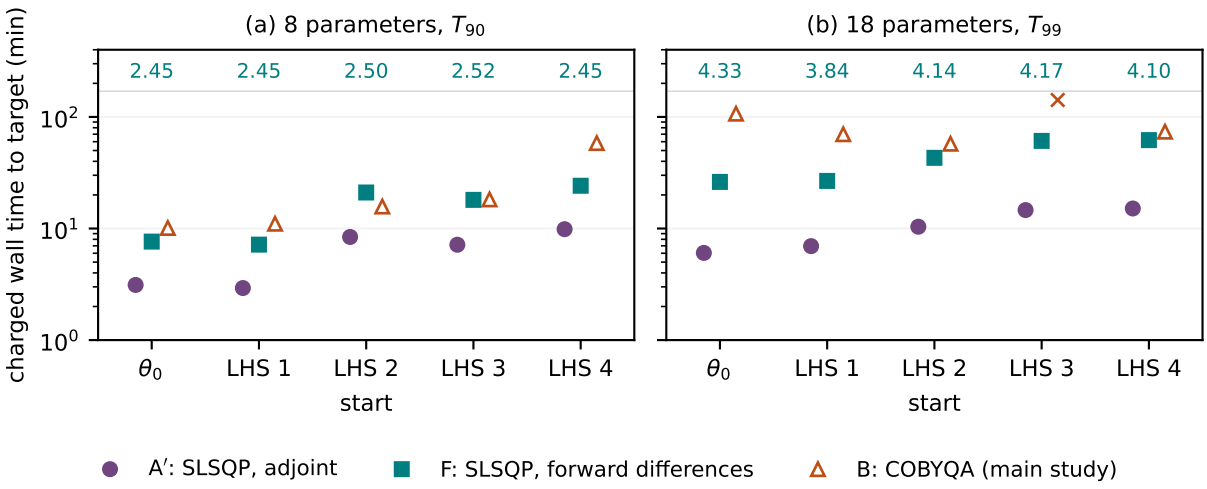


Figure 5: Charged wall time, in minutes, to the primary target from each start (repeat 1): A′ (SLSQP with adjoint gradients) and F (the same SLSQP with forward-difference gradients) from the ablation session, and B (COBYQA) from the main study at the same starts. Minutes avoid the 6% difference between the two sessions' time units. The cross marks the start from which B did not reach T_{99}; it is drawn at B's 120-unit budget. The row of numbers above the line gives each start's F/A′ ratio of charged time, the estimand of Table 8.

finite-difference gradient costs F a median of 1.90 units with eight parameters and 4.25 with 18, about 0.24 units per stencil solve, while one adjoint gradient costs A′ 0.057 and 0.061 units: per gradient, F pays 33 and 70 times as much. Two factors shrink that ratio. Every training evaluation, with or without a gradient, costs a training solve that both arms pay (medians 0.240 and 0.236 units for F, 0.232 and 0.245 for A′), so an evaluation with a gradient costs F 7.4 and 14.7 times as much as A′. And only a median 68% and 77% of F's evaluations request a gradient, the others being line-search evaluations, whereas A′ requests one at every evaluation; with both factors the expected ratio is 5.3 and 11.5, against the measured 5.58 and 10.07 once validation is excluded.

In space B all 20 runs stop at the T_{99} crossing. After the crossings in space A, F ran to the budget in eight of ten runs, without improving on A′'s best designs (Appendix D).

Gradient accuracy. On the L1 training model, with L2 validation as in the session and the frozen step, F's gradients agree with the adjoint within a max-norm relative error of 5.2×10^{-4} to 1.9×10^{-3} for F and at most 7.2×10^{-4} for the full-current mean at θ_0 and an interior point in each space, but one small component exceeds the check's per-component limit of 10^{-1} (Appendix D). A step sweep at that point shows the excess to be forward-difference truncation error: a central difference with the same step reduces it from 0.24 to 2.2×10^{-4} relative. A′ reproduces the main-study A within 6% (Appendix D).

The optimizer factor. Against the main study, B/A = (B/F)(F/A′) splits the SLSQP-versus-COBYQA ratio approximately into an adjoint-versus-finite-difference factor and a factor for COBYQA against SLSQP driven by finite differences. Over the starts where B attains the primary target (all five in space A; four in space B, where B misses LHS 3), the main-study B/A geometric means are 3.10 and 7.79, and F/A′ over the same starts is 2.47 and 4.10. B/F per start, in each session's units, is 1.24, 1.45, 0.70, 0.95 and 2.27 in space A (geometric mean 1.22) and 3.84, 2.46, 1.26 and 1.12 in space B (1.91); in seconds, which removes the 6% difference in time unit, the means

are 1.30 and 2.03. The chain closes only approximately across sessions: (F/A′)(B/F) is 3.02 and 7.82 in units against 3.10 and 7.79. The adjoint factor is nearly the same from every start, whereas the optimizer factor varies from start to start: with eight parameters COBYQA matched or beat finite-difference SLSQP from two of five starts (B/F 0.70 and 0.95 in units, 0.75 and 1.01 in seconds), and with 18 parameters finite-difference SLSQP was faster from every start. The final checks of F's and A′'s selected designs keep all ten target crossings within their targets at L3, with at least 15% margin (Appendix D).

Warm-started stencils (method F_w). F's cold stencil solves make it a weaker competitor than a finite-difference SLSQP that starts each stencil solve from the base state, as the derivative timings of Appendix H do at 0.30–0.40 of a cold forward per perturbed sweep. A second post-hoc arm, F_w, added after the outcomes of the main study and of F were known, measures that competitor under a public protocol frozen before any of its runs. It is F except that each stencil solve starts Newton from the base point's converged training state on the same mesh generation, and it runs with an in-session A′ control from the five starts, repeat 1 only, after an L1 accuracy check of the warm-started stencils. Per-start F_w/A′ ratios: [PENDING exp2c-fd-warm]. F_w against F, across sessions: [PENDING exp2c-fd-warm]. Per-gradient cost and warm against cold stencil-solve cost: [PENDING exp2c-fd-warm]. L1 accuracy of warm-started stencils: [PENDING exp2c-fd-warm]. Final checks of F_w's selected designs: [PENDING exp2c-fd-warm].

6 Design-gradient verification

6.1 Differentiated model and excitation policy

For a design vector p, the converged state z contains the reduced potential and, for harmonic coupling, its multipliers. Writing the full constrained residual as $R(z, p) = 0$, the discrete adjoint is

$$R_z^T \psi = F_z^T, \qquad \frac{\mathrm{d}F}{\mathrm{d}p} = F_p - \psi^T R_p. \tag{3}$$

We differentiate mesh coordinates, element terms, magnet sources, current loads, harmonic constraints and the torque functional. The explicit coordinate derivative of Arkkio is included for conforming coupling. Fixed-topology mesh motion $X(p)$ supplies the chain rule $\mathrm{d}F/\,\mathrm{d}p_k = \langle \mathrm{d}F/\,\mathrm{d}X, \partial X/\partial p_k \rangle$ for geometric parameters, with direct terms added where needed.

The 12 parameters comprise nine geometric lengths (MAG, MH, MW, SW1–SW4, SD1 and Sr1), axial length, operating angle and radial scale. The two designs are the initial parameters and an interior point halfway toward the saved optimized scalar parameters (Appendix J). This verifies a local scalar parametrization. The reference's optimized geometry also uses grouped surface-control displacements; reproducing its 12 scalar values alone does not reproduce that geometry. The gradients that the optimizers of Section 4 used, including the grouped offsets of space B and its mesh-motion map, are checked separately there.

The excitation rule changes these derivatives. Holding peak current fixed differs from the reference optimizer's rule, which holds winding-group-1 current density fixed (Appendix H defines both). At L1 (harmonic coupling, coupling torque, full current), the mean-torque derivative with respect to the stator tooth width SW1 is $+1.68\,\mathrm{N\,m/m}$ under fixed current and $-220.85\,\mathrm{N\,m/m}$ under fixed group-1 density (Fig. 6): a wider tooth narrows the slot and lowers current under the density rule. At this state the policy changes seven of the ten geometric/scale derivatives and reverses the mean-torque signs for SW1, SW4 and SD1. The fixed-size problem of Section 4 holds physical current fixed for every method.

6.2 Finite differences and Taylor remainders

The derivative objectives are mean torque and population standard deviation on the ten even mechanical angles, matching the reference's optimization grid. Central differences move the same mesh at $p \pm he_k$; step sweeps separate truncation from floating-point and nonlinear-solve error. We also check the first-order remainder $|F(p + he_k) - F(p) - h\ \mathrm{d}F/\ \mathrm{d}p_k|$ for quadratic decay.

Across initial and perturbed designs, L0–L2 and the recorded loaded current/rule subsets, there are 800 objective–parameter records. They pool harmonic/coupling-torque and conforming/Arkkio evaluations. The fixed-density MAG, MH and MW records repeat 96 fixed-current comparisons byte-for-byte, leaving 704 distinct comparisons. Of these, 688 have a best-step relative error at most 2.8×10^{-7}; the maximum is 1.07×10^{-5} for a small Sr1 ripple derivative. This is a best-step verification statistic, not the accuracy of a finite-difference step chosen in advance. Figure 6 includes the 704 distinct cases and an explicit Taylor sweep. The artifact retains every step, rejected displacement and relative error. The recorded subsets do not form a full factorial study. Of 592 eligible distinct loaded comparisons, 530 have a quadratic Taylor-remainder window spanning at least four decades of step size. Verified is not the same as model-independent: at the same design the two model–estimator pairs give ripple sensitivities whose box-scaled vectors differ by 6.7% and 10.2% of their norm, more than either changes under mesh refinement. Appendix H gives the Taylor windows, the weaker zero-current checks, mesh convergence and this model dependence.

On one loaded host, across L0–L2, both couplings and tolerances 10^{-8} and 10^{-14}, the full gradient of both aggregate objectives with respect to all 12 parameters costs 1.13–1.20 forwards, whereas central differences cost 8.2–10.5 including the shared base solve (Fig. 6d; Appendix H gives the protocol and seconds).

7 What transfers, and limits

Three findings should transfer beyond this motor. First, with the path held fixed, the adjoint's advantage over finite differences is a ratio of gradient costs, which scales with the number of parameters and holds for the finite-difference configuration measured: forward differences whose stencil solves run serially, one core per run, and start Newton-cold. Solving the stencil in parallel would shorten the wall time per gradient but not its compute; warm-starting it would lower both [PENDING exp2c-fd-warm]. The measured ratios are smaller than the per-gradient cost ratio because both arms pay a training solve at every evaluation, F requests a gradient at only some of them, and validation is shared. Second, the gradient-based optimizer does not pay by itself: without the adjoint, SLSQP was not reliably faster than COBYQA in the eight-parameter space, and its lead in the 18-parameter space was about a factor of two (Section 5). Third, validating each incumbent compresses same-resolution ratios toward one and inflates SLSQP's lead over the coarse competitor D, which needs more validated incumbents (Section 4).

The evidence has clear limits. It is a paired case study over five deterministic starts of one motor; the ratios are not population estimates. The ablation is post hoc, and F's forward differences carry a truncation error that exceeds the per-component check limit on one small component (Section 5). CMA-ES and other population or multi-start methods remain untested. Method C ran with the preregistered settings of Appendix B and was not tuned; its low attainment bounds this configuration, not radial-basis surrogates in general. No reduced physical model, such as a nonlinear magnetic equivalent circuit (Kemmetmüller et al., 2015), a waveform-targeting surrogate (Farshbaf Roomi et al., 2022) or an adaptive reduced model (Wen and Zahr, 2023), was compared: D is a coarse finite-element model, not such a baseline, and the slotless screening control of Appendix I cautions against inferring an advantage from parameter count alone. In space A the parameterization and the active mean floor, more than the method, set the attainable ripple: four of five methods reach

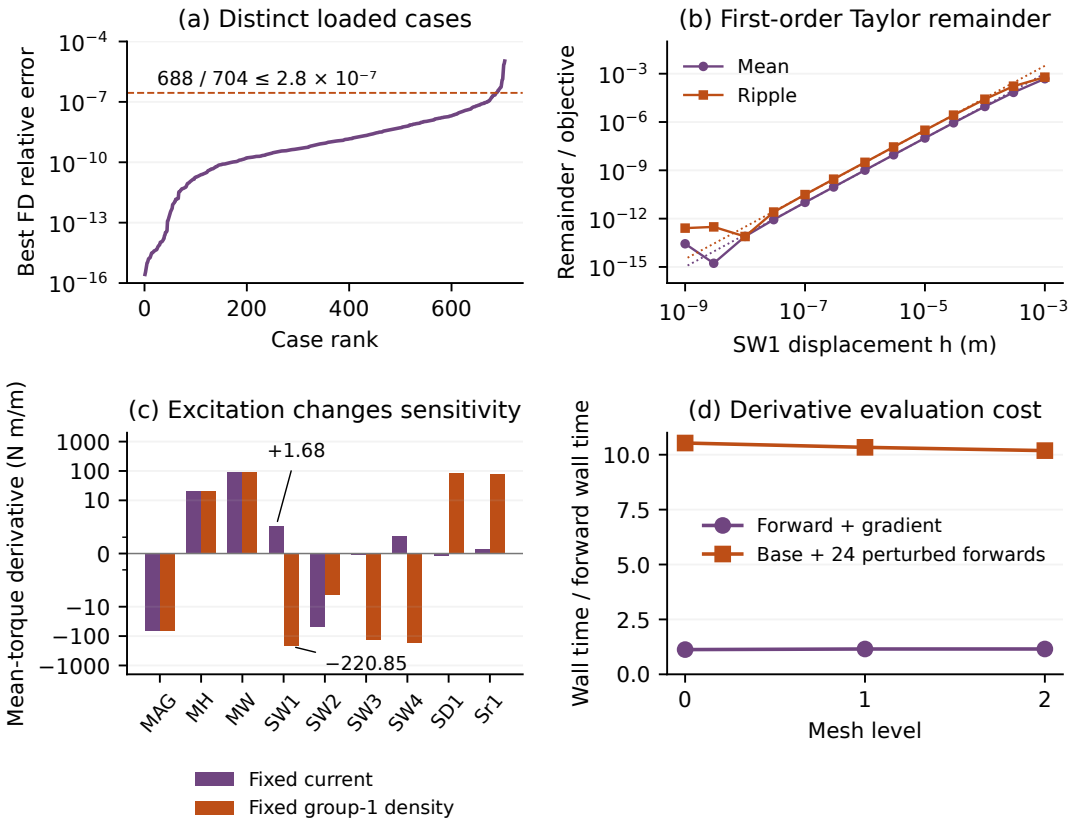


Figure 6: Discrete gradient evidence. (a) The 704 distinct loaded best-step finite-difference errors, sorted, pooling harmonic/coupling-torque and conforming/Arkkio objective–parameter cases. (b) SW1 Taylor remainders at L1, initial design, full current, harmonic coupling and coupling torque, fixed current; the plotted remainders are divided by $|F(p)|$, and dotted h^2 guides are anchored at $h = 10^{-5}$ m. (c) Mean-torque sensitivities for the nine geometric parameters under two current rules at that same state; the vertical axis is symmetric-log, linear within ± 2 N m/m. (d) Recorded cost ratios for harmonic coupling at tolerance 10^{-14}; gradients include the forward evaluation, and FD includes its shared base forward. L0/L1 use medians of three runs, L2 a single run; the seconds are in Appendix H. These timing data were collected under uncontrolled host load.

the same $0.043F_0$ (C reaches $0.049F_0$). The forward model is two-dimensional and magnetostatic, without thermal, demagnetization, fatigue, end-effect or loss models, and is compared with published computational artifacts, not with measurements on a physical motor.

8 Conclusion

Before trusting a motor gradient, check what it differentiates: interface order and excitation rule change the objective or its sensitivities. On a common validation model, SLSQP with adjoint gradients beat derivative-free COBYQA in every comparison where both succeeded. A post-hoc ablation that kept SLSQP but used forward differences took 2.5 and 4.1 times as long with 8 and 18 parameters on one core per run with cold-started stencils, so the adjoint, more than the gradient-based optimizer, accounts for that advantage [PENDING exp2c-fd-warm].

Code and data availability

The original forward and gradient evidence is fixed at GEODE commit `910593ba`. The repository contains the raw motor runs and analysis, geometry and mesh-generation scripts, design-gradient sweeps and timing records, and solver verification logs. Candidate-validation records are pinned at `224332c5`, and independent reference reruns at `8acd21ec`. Reference data are extracted from the authors' ICEM snapshot at `b145aaf4`, a later repository revision rather than a claim to the original conference release. Source hashes, extraction checks and the upstream GPL-3.0-or-later license are retained. Recomputing the physical study requires the separately documented mesh and solver commands; the benchmark READMEs record the GEODE-side Gmsh and Rust toolchain versions.

The matched optimization study (Section 4) follows the optimization harness at `9486a9da`, the source of the study's evaluator, and its training-gradient finite-difference records; the final-refinement program is at `3e4ee0c6`. The paper package is the directory `papers/motor-gradient-optimization/motor-gradient-optimization.14` of the GEODE repository. It contains a manifest of every copied input with its source commit and SHA-256, the figure inputs and regeneration scripts (rebuilding the figures does not require rerunning FEM), three standard-library verifiers that recompute the reported numbers from those inputs, the per-run validation histories and ledger charges, the 213 final-check results, copies of the training-gradient checks, and the study's SHA-256 inventory of all 5,453 output files. The inventory was taken when the session finished; its existence by 2026-10-20 03:50 UTC is timestamped (RFC 3161 and OpenTimestamps), and each extracted run and session record is bound to it by hash. The session, calibration and anchor-recomputation records are included, as is an export of the governing preregistration comments taken on 2026-10-18 at 09:34 UTC, before the main session began at 18:16 UTC.

For the post-hoc ablation, the package holds the 40 run extracts, the setup records, the ablation's SHA-256 inventory of 2,118 files with its RFC 3161 and OpenTimestamps records, the final-check results, an export of its protocol; the ablation's evaluator source is `5cc86ad8`; F's L1 gradient check is pinned at `74843242` and its step sweep at `f94acad3`. The ablation's protocol was posted at 22:34 UTC on 2026-10-20 and frozen by a comment at 23:00 UTC, before the first ablation run began at 02:07 UTC on 2026-10-21; these are GitHub timestamps, and the protocol text was never edited. The raw traces of both studies, the 5,453 and 2,118 inventory-listed files with both inventories and their timestamp records, are packed in one archive, `motor-paper-raw-traces.tar.zst` (2,179,756,244 bytes, SHA-256 `2de2132cb6b94c768ace042ad8cd85051000aee148d520cd1843d1c10d40b2a2`), for deposit at DOI [PENDING zenodo-doi]. Every file was checked against its inventory when the archive was built; given the unpacked ablation session, the package's third verifier re-derives the ablation extracts from the raw traces.

A Field model details

A.1 Nonlinear field equation

The two-dimensional magnetic vector potential $A = A_z$ determines the flux density $\mathbf{B} = (\partial_y A, -\partial_x A)$. Iron is described by an isotropic reluctivity $\nu(|\mathbf{B}|^2)$, where reluctivity is the inverse of permeability. For a test function v that satisfies the homogeneous constraints, the weak equation is

$$\int_\Omega \nu(|\nabla A|^2)\nabla A \cdot \nabla v \,\mathrm{d}\Omega = \int_\Omega J_z v \,\mathrm{d}\Omega + \int_{\Omega_m} \mathbf{M} \cdot (\partial_y v, -\partial_x v)\,\mathrm{d}\Omega. \tag{4}$$

Here $\mathbf{M}$ denotes the permanent-magnet source in $\mathbf{H} = \nu\mathbf{B} - \mathbf{M}$. In the magnet, $\mu_r = 1.05$ and $B_r = 1.4\,\mathrm{T}$, so its source magnitude is $H_c = B_r/(\mu_0\mu_r)$, not B_r/μ_0. Its direction is $\pi/2$ in the rotor frame and rotates with the rotor. Air and windings have $\mu_r = 1$. The inner and outer circular boundaries have $A = 0$.

We use continuous piecewise-linear triangles. With nodal vector u, element gradients $g_i = \nabla\varphi_i$, $G = \nabla A$ and $s = G \cdot G$, the residual derivative on one element is

$$K^e_{ij} = |\Omega_e| \left[\nu(s) g_i \cdot g_j + 2\nu'(s)(g_i \cdot G)(g_j \cdot G)\right]. \tag{5}$$

The second term is required when the material changes with field strength. Newton steps use this consistent tangent, with damping and a stopping reference independent of the initial guess. Constant potential offsets, where allowed by the boundary conditions, do not change that reference. The solver reports whether it stopped at the requested tolerance or an estimated floating-point floor. All 500 baseline and sensitivity solves, and all 1,140 additional harmonic-coupling solves, stopped at the requested tolerance.

The iron law uses 47 tabulated M27 B–H points from 0 to 2.30 T. Monotonicity is imposed on $H(B)$, not on $\nu(B^2) = H(B)/B$. The tabulated H/B is non-monotone up to about 0.75 T. The two material tangent eigenvalues must satisfy

$$\nu > 0, \qquad \nu + 2s\nu' = \frac{\mathrm{d}H}{\mathrm{d}B} > 0. \tag{6}$$

The GEODE and reference interpolants agree to round-off between 0.05 and 2.25 T. GEODE uses an odd cubic near zero and a continuous slope into the vacuum-slope tail above 2.30 T. The reference uses a low-field clamp and a different last-knot slope. The maximum relative difference in reluctivity near zero is 6.66%; the last segment differs in H by at most 2.17×10^{-4} relatively. Appendix F tests the torque effect of the low-field difference.

A.2 Rotation, coupling and torque

At the modeled angles, the two circular traces have matching nodes after rotation. Our conforming interface identifies their potentials, including the antiperiodic sign on sector wrap-around. Writing the full nodal vector as $u = P\hat{u} + u_D$, the reduced residual and tangent are $\hat{R} = P^T R(P\hat{u} + u_D)$ and $\hat{K} = P^T K P$. The rotor geometry and magnet source rotate together; currents use the same mechanical angle in Eq. (1).

We average Maxwell stress over the air gap using the Arkkio expression (Arkkio, 1987)

$$T(\beta) = \frac{L}{\mu_0(r_2 - r_1)} \int_{\Omega_{\mathrm{gap}}} r B_r B_\theta \,\mathrm{d}\Omega, \qquad (r_1, r_2) = (44, 45)\ \mathrm{mm}. \tag{7}$$

Positive torque acts counter-clockwise on the rotor. All torques in this paper include axial length and refer to the modeled sector. We apply no factor of six when comparing with the saved reference arrays, despite a per-length label appearing in the reference evaluation plots.

The alternative harmonic model imposes

$$\int_{\Gamma_c} (A_s - A_r) e_n \,\mathrm{d}\Gamma = 0, \qquad e_n \in \{\sin(n\theta), \cos(n\theta)\}, \quad n = 3, 9, \ldots, N_h, \tag{8}$$

with maximum order $N_h = 45$ unless stated otherwise. The 16 order-45 constraints match the reference's eight orders. Rotor and stator retain distinct trace unknowns. With constraint matrix C, Lagrange multipliers η and reduced potential $\hat{u}$, Newton's method uses the saddle tangent

$$\begin{bmatrix} K & C \\ C^T & 0 \end{bmatrix}. \tag{9}$$

The trace moments use the exact Fourier integrals of the piecewise-linear hats. Rotation enters their centers; the constrained energy derivative gives

$$T_c = -L\eta^T \frac{\partial (C^T \hat{u})}{\partial \beta}, \tag{10}$$

with the state held fixed in the partial derivative, consistent with Egger et al. (2022). Forward comparisons use Arkkio for both GEODE couplings to isolate the interface change; coupling torque is a separate check. The derivative study uses coupling torque for the harmonic model and Arkkio for the conforming model. The two objectives and their gradients are consequently comparisons of those specified model–estimator pairs. The optimization study trains on order-45 coupling torque and validates on order 93.

The main harmonic runs retain antiperiodic trace-end pairing. An additional set frees those ends, matching the reference's treatment. The full-order identity check is summarized in Section 3.

A.3 Implementation checks

The nonlinear implementation is checked against linear assembly, residual finite differences, a manufactured solution and a saturated annulus with an independent radial field relation. Sector tests compare the constrained solution with a full-machine solve, check the reduced tangent by finite differences, and exercise positive, negative and wrapped rotation steps. Clockwise and counterclockwise triangle orderings give equivalent magnet sources and fields. The motor regression also detects changes in magnet strength, magnet direction, phase ordering and rotation direction. The verification records are linked in the data-availability section.

B Optimizer and search settings

Table 3 lists the settings a replication needs beyond Section 4; the pinned harness documents them in full. All methods work in the unit box $u \in [0,1]^n$ of the free coordinates, minimize $F_{\text{train}}/F_{\text{train}}(\theta_0)$ and impose the mean floor as $\mu_{\text{train}}/\mu_{\text{train}}(\theta_0) - 1.0025 \geq 0$. No method validates its start point, and a point validated once is not validated again.

Table 3: Optimizer settings. n is the number of free coordinates (8 or 18), x_v the last admissible evaluated point and x a refused request.

Method	Settings
A, E	SciPy SLSQP, `ftol` 10^{-15}, `maxiter` 10^5. Adjoint gradients of the objective and the mean constraint; analytic Jacobians of the geometry constraints. Safeguard: a refused request (unbuildable, mesh-infeasible or derivative undefined) aborts the current solve; the first admissible point $y_k = x_v + 2^{-k}(x - x_v)$, $k = 1, \ldots, 20$, is restored and SLSQP restarts from it with fresh state, at most 20 times. Validation: each iterate the SLSQP callback reports.
B, D	SciPy COBYQA, $2n + 1$ initial interpolation points, trust radius 0.1 to 10^{-6}, `maxiter` = `maxfev` = 10^9. A refused point returns NaN for objective and mean constraint; geometry constraints keep their values. Validation: the callback's best point whenever it changes, after the initial set.
C	Four cubic radial-basis interpolants, $\phi(r) = r^3$ with a linear tail: the three training-current standard deviations, each over $\sqrt{F_{\text{train}}(\theta_0)}$, and the full-current mean, refitted after every evaluation. Initial design: the start and $u_0 \pm \Delta_0 e_i$, $\Delta_0 = 0.1$ ($n + 1$ points). Subproblem: SLSQP on the surrogate in the box and $\|u - u_k\|_\infty \le \Delta$, at most 200 iterations. Acceptance: a candidate better than the incumbent, feasible first (lower violation, then lower F). On acceptance with $\rho \ge 0.75$ and a step of at least 0.9Δ, $\Delta \leftarrow \min(2\Delta, 0.5)$; on rejection $\Delta \leftarrow \Delta/2$, with a geometry point $u_k \pm \Delta e_j$ after every second rejection. Stop at $\Delta < 10^{-6}$. Refused points are skipped and count as rejections. Validation: each acceptance.

The Latin-hypercube starts (McKay et al., 1979) use seeds 1–4; the computations use NumPy (Harris et al., 2020) and SciPy (Virtanen et al., 2020).

C Study protocol details

Training gradients. Section 6 verifies a different, scalar parameter set; the gradients that A and E use were checked separately. In both spaces, the L1 training objective F and full-current mean were compared with central differences at θ_0 and at an interior point, along a fixed box-scaled direction and for three (space A) or four (space B) components, including the offsets CPR1 and CPS1. With Newton tolerance 10^{-12}, the directional relative errors at $h = 10^{-4}$ are 1.5×10^{-8} to 2.8×10^{-7} for F and 2.4×10^{-9} to 4.7×10^{-9} for the mean, and the component errors at $h = 10^{-6}$ are at most 2.8×10^{-6}. The directional error falls by 72 to 156 times from $h = 10^{-3}$ to 10^{-4}, as truncation error should. The evaluator's own gradient replies at the study's training tolerance of 10^{-8} agree at the interior point within 3.1×10^{-6} directionally; the largest component error, 4.3×10^{-5}, belongs to the small Sr1 derivative. At θ_0, space B's offsets sit on their bounds, so the central stencils there step outside the design box. The records are pinned with the optimization harness (Code and data availability).

Session. The session ran on one AWS c7a.4xlarge host (x86-64; NumPy 2.5.4 and SciPy 1.18.1 with OpenBLAS) with 15 concurrent single-threaded processes. Before calibration, the host recomputed the initial-design validation waveforms in both spaces. F_0 agreed with the canonical anchor to 2.4×10^{-13} relative, within the preregistered 10^{-9} tolerance, and the two spaces agreed bit for bit. The calibrated unit was 70.94 s, the median of 45 concurrent cold validations. All 100 runs completed in one session with no recovery, deferral or contamination: the one-minute load average peaked at 21.1, below the contamination threshold of $1.5N = 22.5$. The in-run cost totals 109 CPU-hours. Of the 100 runs, 38 stopped on reaching T_{99}, 46 on optimizer convergence and 13 at the budget. The

other three are SLSQP runs (A) in space A whose solver stopped without converging: both repeats from LHS 2, after about 43 units, and repeat 1 from LHS 3, after 119 units.

Refinement host. The registered pipeline begins by capturing the closed study on its original host, a step that could not be completed after that host had been restarted. The refinement instead used a copy of the study outputs whose integrity is established by a SHA-256 inventory of every output file, taken when the session finished, and independently timestamped; the timestamps establish that the inventory existed by 2026-10-20 03:50 UTC. It ran on a different host (Apple M3 Ultra, macOS arm64). For the 16 selected designs validated on the moved reference mesh, the primary check reproduces the study's F_{val} to below 2.2×10^{-12} relative. The other 15 candidates needed a fresh mesh, which Gmsh triangulates differently on the two platforms; their F differs by at most 1.7×10^{-3} relative. The remaining two designs are θ_0 in each space, which reproduces F_0. The fresh-mesh differences measure the sensitivity of F to remeshing and are far below the 10% allowance δF.

D Per-start times, ratios and final checks

Table 4: Charged wall-time units to the first attaining validation, repeat 1. A dash means the target was not attained within 120 units. T_{99} was attained by no run in space A.

Space, target	Method	θ_0	LHS 1	LHS 2	LHS 3	LHS 4
A, T_{50}	A	2.7	2.5	2.6	2.7	5.1
	B	8.5	9.3	9.3	8.9	13.7
	E	1.9	1.8	1.0	3.0	—
	D	3.1	5.3	7.3	4.8	8.8
	C	10.0	9.8	4.1	10.6	—
A, T_{90}	A	2.7	2.5	7.4	6.4	8.8
	B	8.5	9.3	13.3	15.4	49.3
	E	1.9	1.8	4.5	3.0	—
	D	4.6	5.3	7.3	9.4	21.9
	C	16.9	9.8	—	—	—
B, T_{50}	A	1.6	1.7	3.0	4.8	1.6
	B	12.2	13.1	13.5	21.5	15.7
	E	1.0	1.1	1.1	4.4	1.2
	D	4.1	6.9	5.2	11.9	8.4
	C	7.3	8.4	7.3	12.1	8.9
B, T_{90}	A	3.1	5.4	3.0	12.0	1.6
	B	14.4	13.1	13.5	36.6	35.3
	E	2.0	—	2.8	10.9	1.2
	D	5.9	8.1	5.2	15.9	14.3
	C	8.4	8.4	10.3	—	14.6
B, T_{99}	A	5.4	6.6	9.1	13.1	13.4
	B	90.4	59.1	48.6	—	62.3
	E	3.9	—	6.5	10.9	8.3
	D	43.0	14.1	36.0	83.3	47.8
	C	—	—	—	—	25.9

Table 5: Paired per-start time ratios (repeat 1). B/A and D/E vary the optimizer at fixed resolution; E/A and D/B vary the resolution; D/A compares A with the coarse derivative-free competitor. Ratios above one favor the denominator. Geometric means use the finite pairs only, so the D/E and E/A means omit the starts where E failed to attain the target.

Space, target	Ratio	Finite pairs	Geometric mean	Per-start ratios
A, T_{50}	B/A	5	3.25	3.18, 3.66, 3.63, 3.25, 2.66
	D/E	4	2.75	1.60, 2.90, 7.68, 1.61, —
	E/A	4	0.68	0.72, 0.72, 0.37, 1.09, —
	D/B	5	0.56	0.36, 0.57, 0.78, 0.54, 0.64
	D/A	5	1.83	1.15, 2.09, 2.85, 1.75, 1.71
A, T_{90}	B/A	5	3.10	3.18, 3.66, 1.80, 2.43, 5.61
	D/E	4	2.45	2.41, 2.90, 1.61, 3.17, —
	E/A	4	0.62	0.72, 0.72, 0.61, 0.47, —
	D/B	5	0.54	0.54, 0.57, 0.55, 0.61, 0.44
	D/A	5	1.68	1.72, 2.09, 0.99, 1.48, 2.49
B, T_{50}	B/A	5	6.49	7.76, 7.70, 4.51, 4.46, 9.59
	D/E	5	4.69	3.90, 6.14, 4.93, 2.72, 7.09
	E/A	5	0.63	0.66, 0.66, 0.35, 0.91, 0.72
	D/B	5	0.46	0.33, 0.52, 0.39, 0.56, 0.53
	D/A	5	2.97	2.58, 4.04, 1.75, 2.48, 5.09
B, T_{90}	B/A	5	5.06	4.64, 2.41, 4.51, 3.05, 21.54
	D/E	4	3.14	2.94, —, 1.86, 1.46, 12.17
	E/A	4	0.79	0.65, —, 0.94, 0.91, 0.72
	D/B	5	0.45	0.41, 0.62, 0.39, 0.44, 0.41
	D/A	5	2.25	1.91, 1.50, 1.75, 1.33, 8.74
B, T_{99}	B/A	4	7.79	16.65, 8.92, 5.33, —, 4.65
	D/E	4	7.20	11.05, —, 5.50, 7.65, 5.79
	E/A	4	0.72	0.72, —, 0.72, 0.83, 0.61
	D/B	4	0.50	0.48, 0.24, 0.74, —, 0.77
	D/A	5	4.32	7.91, 2.12, 3.94, 6.36, 3.56

Table 6: Final checks of the selected designs. "best" marks a method's representative; "A T_{90}" marks the earliest repeat-1 crossing of that target by that method. The L2 column is the primary check on the refinement host. L3 and conforming changes are in percent; the 40-angle, tight and shifted changes are relative, in units of 10^{-4}. "held-out" is the objective at currents 0.25 and 0.75 relative to θ_0's at the same currents. A dash means the conforming check does not apply because stator offsets are nonzero.

Design	F_{L2}/F_0	L3 (%)	g40 (10^{-4})	tight (10^{-4})	shifted (10^{-4})	conf. (%)	held-out
Space A							
θ_0	1.0000	−1.3	0.0	0.0	+0.1	+2.1	1.0000
best A	0.0432	−0.4	0.0	0.0	+0.1	+4.4	0.0472
best B	0.0432	−0.5	0.0	0.0	0.0	+4.4	0.0472
best C	0.0489	−0.7	+0.2	0.0	+0.5	+4.1	0.0524
best D	0.0432	−0.5	0.0	0.0	0.0	+4.4	0.0472
best E	0.0434	−0.6	+0.4	0.0	+0.9	+4.4	0.0473
A T_{50}, A T_{90}	0.0653	−2.9	+0.7	0.0	+1.5	+1.1	0.0686
B T_{50}, B T_{90}	0.0881	−2.3	+1.3	0.0	+2.6	+3.2	0.0866
C T_{50}	0.1472	−4.0	+0.6	0.0	+1.3	+3.3	0.1508
C T_{90}	0.0507	−0.9	+0.2	0.0	+0.5	+4.0	0.0543
D T_{50}	0.0920	−2.2	+1.5	0.0	+3.0	+3.3	0.0899
D T_{90}	0.0594	−2.3	+1.3	0.0	+2.6	+1.6	0.0609
E T_{50}	0.1105	−2.0	+0.6	0.0	+1.1	+1.2	0.1104
E T_{90}	0.0660	−2.9	+0.7	0.0	+1.5	+1.1	0.0694
Space B							
θ_0	1.0000	−1.3	0.0	0.0	+0.1	+2.0	1.0000
best A	0.0053	+2.6	+0.4	+0.1	+0.5	—	0.0066
best B, B T_{99}	0.0085	−3.0	−0.4	0.0	−1.2	—	0.0076
best C, C T_{99}	0.0057	−3.7	+2.5	0.0	+5.0	—	0.0074
best D	0.0044	+6.3	+2.6	+0.1	+4.9	—	0.0076
best E	0.0071	+2.6	−1.5	0.0	−3.1	—	0.0087
A T_{50}	0.1874	+0.9	−0.3	0.0	−0.6	—	0.1538
A T_{90}	0.0255	+1.1	−0.1	0.0	−0.2	—	0.0248
A T_{99}	0.0077	+0.4	−0.1	0.0	−0.3	—	0.0096
B T_{50}	0.1854	−1.5	−0.1	0.0	−0.3	—	0.1907
B T_{90}	0.0523	−0.4	−0.2	0.0	−0.4	—	0.0505
C T_{50}	0.3958	−1.4	−0.1	0.0	−0.1	—	0.3692
C T_{90}	0.0259	+0.7	0.0	0.0	0.0	—	0.0247
D T_{50}	0.1925	−1.5	−0.1	0.0	−0.3	—	0.1971
D T_{90}	0.0393	−0.7	+0.6	0.0	+1.2	—	0.0392
D T_{99}	0.0089	+0.4	−0.5	0.0	−1.0	—	0.0137
E T_{50}	0.1961	+0.8	0.0	0.0	0.0	—	0.1567
E T_{90}	0.0223	−2.8	−5.3	0.0	−10.6	—	0.0144
E T_{99}	0.0081	+0.7	+0.2	0.0	+0.4	—	0.0112

Held-out currents. The held-out currents test the ripple reduction outside the three training currents. We compare each design's objective at currents 0.25 and 0.75, relative to θ_0's at the same currents, with its relative objective at the training currents. Across the candidates, the held-out ratio is 0.65 to 1.73 times the training-current ratio; values above one mean a smaller reduction at the held-out currents. For the five best space-A designs it is 1.07–1.09. In space B, four of the five best designs keep less of their reduction: their remaining objective fraction at the held-out currents is 1.23–1.73 times that at the training currents, while for B's best design it is 0.89. D's best design has $0.0044F_0$ at the training currents but 0.0076 of θ_0's value at the held-out currents. Part of the extra reduction from space B's added freedom is therefore specific to the three training currents.

Validation in the ablation. Validation is a median 61–65% of A′'s time to a target and 16–27% of F's (Table 9), which pulls the measured ratio toward one.

Ablation runs after the crossings. After the crossings in space A, F's SLSQP met its stopping test in two of ten runs and ran to the budget in the other eight, whereas A′ converged in four,

Table 7: Validation charges, repeat 1. "To crossing" counts the validations up to and including the first attaining one, and "share" gives the fraction of the time to that target charged to validation, as medians (ranges) over the method's attained targets and starts. "Run" gives the range over starts of the fraction of the whole run's charged time spent in validation.

	Space A			Space B		
Method	To crossing	Share	Run	To crossing	Share	Run
A	2 (2–7)	0.65 (0.65–0.71)	0.45–0.71	3 (1–10)	0.61 (0.58–0.68)	0.65–0.68
B	5 (2–25)	0.46 (0.22–0.53)	0.24–0.34	8 (2–36)	0.34 (0.15–0.45)	0.35–0.49
C	6.5 (2–10)	0.53 (0.42–0.64)	0.02–0.63	4 (2–17)	0.40 (0.25–0.56)	0.50–0.61
D	6 (2–20)	0.81 (0.60–0.84)	0.41–0.64	9 (2–70)	0.71 (0.46–0.77)	0.73–0.77
E	2 (1–5)	0.90 (0.87–0.92)	0.82–0.90	3 (1–11)	0.88 (0.83–0.90)	0.87–0.90

stopped in the solver in four and reached the budget in two. The extra time bought F nothing: from every start its best validated objective is within 0.4% of A′'s. Why F rarely met the stopping test (`ftol` 10^{-15}, Table 3) was not tested; it does not affect the time to target, since every crossing happens at the same validation count.

Control. As a check of the control, A′'s primary crossing times are 0.94–1.04 times those of the main-study A from the same start and repeat, each in its own session's unit.

Training and validation objectives. The training model also reads lower than validation. At θ_0, the order-45 training objective is 0.887 (L1) and 0.894 (L0) of F_0. At the 31 selected designs, the ratio of training to validation objective lies between 0.57 and 1.19 times its θ_0 value. The lowest values belong to space-B designs near T_{99}: E's and A's T_{99} crossings (0.57 and 0.71) and D's best design (0.65); the best space-B designs of B and C are at 0.99 and 1.19. Large reductions therefore often, but not always, come with a training objective that underestimates validation more than at θ_0. This is consistent with the cancellation conjecture of Section E.2 but does not establish it.

Ablation session and finite-difference accuracy. On the L0 model, F's gradients agree with the adjoint to a max-norm relative error of 5.1×10^{-4} to 1.9×10^{-3} for F and 1.0×10^{-4} to 6.7×10^{-4} for the full-current mean, at θ_0 and an interior point in each space. Single small components are less accurate, up to 5.7×10^{-2} for SW3 at the space-B interior point. These errors, at the frozen step, are part of what F is; the frozen acceptance limits were 5×10^{-3} for the norm and 10^{-1} per component. The last gradient of each space-A θ_0 run of F was cut off by the budget and is excluded from the cost medians. The L1 check was run after the ablation, with the frozen step and the frozen limits; it was not part of the ablation's protocol. Its status is FAIL because of one component: at the space-A interior point, the SW3 component of the mean gradient is -9.62×10^{-3} by finite differences against -7.74×10^{-3} by the adjoint, a relative error of 0.24, on a component 8.7×10^{-5} times the largest. The records show a pattern rather than scatter. For every one of the eight scalar components, the finite-difference-minus-adjoint error of the mean gradient has the same sign at all four points, at L1 and at L0, and for SW3 it is nearly the same at all four (three distinct designs, since θ_0 is shared by both spaces): -1.8 to -1.9×10^{-3} at L1 and -3.0 to -3.1×10^{-3} at L0. The SW3 failure is therefore this offset on a component whose adjoint value is small at that design. A step sweep at the failing point, recorded separately from the L1 check, identifies the offset. For SW3, MH and SW2, forward differences at steps of 10^{-5}, 10^{-6} and 10^{-7} m give an error that, divided by the step, keeps its sign and its size to within 0.9% (3.0% for the MH component of the

objective gradient), as truncation error does and noise would not. Central differences remove it: the SW3 mean component's error falls from 0.24 relative (forward, 10^{-6}) to 2.2×10^{-4} (central, 10^{-6}), and its central error falls 94-fold from 10^{-5} to 10^{-6}, close to the 100-fold that h^2 predicts. At 10^{-6} the remaining central errors of the MH and SW2 components are 9×10^{-8} to 3.4×10^{-7} in absolute terms, so computational noise at that step is of that order or smaller, unless it cancels against the remaining truncation term. For the mean gradient, SW3's forward error per unit step, $1.9 \times 10^3\,\mathrm{N\,m/m^2}$, is smaller than MH's ($2.8 \times 10^3\,\mathrm{N\,m/m^2}$) and SW2's ($9.7 \times 10^3\,\mathrm{N\,m/m^2}$), so SW3 fails because its mean-gradient slope at this design is small ($-7.7 \times 10^{-3}\,\mathrm{N\,m/m}$), not because its curvature is large. In the terms of Moré and Wild (2011, 2012), the frozen step lies where truncation, not computational noise, dominates the forward-difference error. The sweep is deterministic: a repeated base solve agrees bit for bit, and its forward values at 10^{-6} equal those of the L1 check.

The session ran on the same type of host as the main study (c7a.4xlarge, 15 concurrent processes) with a calibrated unit of 66.66 s, against 70.94 s in the main session. The anchor recomputation again agreed within 2.4×10^{-13}, no run was host-contaminated, and the one-minute load peaked at 19.7. A SHA-256 inventory of the 2,118 session files was taken on the host before it stopped and timestamped by two RFC 3161 authorities; every ablation number in this paper is recomputed from traces bound to it.

Ablation final checks. The final checks confirm F's and A′'s selected designs (Table 10). All 90 checks of the 14 designs completed. The primary check reproduces the study's F_{val} within 6.3×10^{-4} relative. L3 changes F by -2.9% to $+2.6\%$ (-2.9% to $+1.1\%$ at the crossings), the 40-angle, tight and shifted checks by less than 0.03%, and the conforming estimator gives 1.1% to 4.4% more ripple on the space-A designs. All ten target crossings keep their target at L3 with at least 15% margin, and every candidate keeps the mean floor, the closest by 0.044%. An independent selection program chose the same 14 designs, bit for bit.

Table 8: Post-hoc ablation (Section 5): charged time units to the first attaining validation for A′ and F, repeat 1, and their ratio per start; the last column gives the geometric mean, with repeat 2 in parentheses. Units are those of the ablation session.

Space, target		θ_0	LHS 1	LHS 2	LHS 3	LHS 4	GM (rep. 2)
A, T_{50}	A′	2.8	2.6	2.7	2.8	5.2	
	F	6.9	6.5	6.5	6.7	12.7	
	F/A′	2.45	2.45	2.39	2.43	2.43	2.43 (2.44)
A, T_{90}	A′	2.8	2.6	7.6	6.4	8.9	
	F	6.9	6.5	18.9	16.3	21.8	
	F/A′	2.45	2.45	2.50	2.52	2.45	2.47 (2.49)
B, T_{50}	A′	1.6	1.6	3.2	4.8	1.7	
	F	6.3	5.6	11.3	17.8	5.9	
	F/A′	3.91	3.59	3.58	3.69	3.60	3.67 (3.68)
B, T_{90}	A′	3.1	5.1	3.2	12.1	1.7	
	F	12.2	20.8	11.3	49.9	5.9	
	F/A′	3.92	4.09	3.58	4.14	3.60	3.86 (3.84)
B, T_{99}	A′	5.4	6.3	9.3	13.2	13.6	
	F	23.6	24.0	38.7	54.9	55.8	
	F/A′	4.33	3.84	4.14	4.17	4.10	4.11 (4.04)

Table 9: Validation charges in the ablation, repeat 1. "To crossing" counts the validations up to and including the first attaining one; "share" is the fraction of the time to that target charged to validation, as medians (ranges) over all attained targets and starts; "run" is the range over starts of the fraction of the whole run's charged time spent in validation.

	Space A			Space B		
Method	To crossing	Share	Run	To crossing	Share	Run
A′	2 (2–7)	0.65 (0.65–0.70)	0.44–0.70	3 (1–10)	0.61 (0.58–0.68)	0.64–0.68
F	2 (2–7)	0.27 (0.26–0.29)	0.25–0.28	3 (1–10)	0.16 (0.15–0.18)	0.15–0.18

Table 10: Final checks of the ablation's selected designs, as in Table 6: L2 objective, L3 change, the largest 40-angle, tight or shifted change, and the conforming change where stator offsets are zero.

Design	F_{L2}/F_0	L3 (%)	max g40/tight/shifted (10^{-4})	conf. (%)
Space A				
θ_0	1.0000	−1.3	0.1	+2.1
A′ T_{50}, T_{90}	0.0653	−2.9	1.5	+1.1
F T_{50}, T_{90}	0.0653	−2.9	1.5	+1.1
best A′	0.0432	−0.4	0.1	+4.4
best F	0.0432	−0.6	0.1	+4.2
Space B				
θ_0	1.0000	−1.3	0.1	+2.0
A′ T_{50}	0.2721	−0.6	3.0	—
A′ T_{90}	0.0255	+1.1	0.2	—
A′ T_{99}	0.0077	+0.4	0.3	—
F T_{50}	0.2720	−0.6	3.0	—
F T_{90}	0.0252	+1.0	0.6	—
F T_{99}	0.0076	+0.4	0.3	—
best A′	0.0053	+2.6	0.5	—
best F	0.0054	+2.5	0.2	—

E Forward reproduction details

Each mesh level contains 60 baseline solves: three current levels and 20 angles. For a scalar statistic q, the estimated L3 refinement interval is

$$q_3 \ \pm \ e_q, \qquad e_q = \max\{|q_2 - q_1|, |q_3 - q_2|\}. \tag{11}$$

We use the same construction pointwise for waveforms. This interval is a refinement diagnostic, not a proven error bound or a statistical confidence interval. It does not cover reference-model error or angular quadrature error. The loaded means change direction between the last two increments, so a last-increment estimate alone would be particularly weak.

Figure 8 in Appendix F shows the improvement angle by angle. The largest full-current difference falls from 0.0272 to 0.0034 N m. The harmonic full-current ripple interval contains the saved reference; the saved half-current value lies approximately 5×10^{-5} N m below its interval. Mean agreement is therefore insufficient to characterize waveform accuracy.

The saved zero-current reference exposes a separate discretization effect. Harmonic coupling changes the cogging standard deviation from 29.1% below to 12.9% above the reference. Its waveform is close to a 1.13-fold scaling of the saved waveform, but the reference lies outside the estimated

ripple interval. This motivates refining the reference itself before interpreting the discrepancy. The harmonic cogging standard deviations are 0.0101, 0.0119, 0.0121 and 0.0121 N m on L0–L3, compared with 0.0107 N m in the reference. The independent reference refinement below tests this coarse-model explanation directly.

Two additional controls narrow the explanation. At L3, coupling torque and the inner- and outer-half Arkkio estimates are each within 9×10^{-5} N m of full-band Arkkio in the order-45 solution. Freeing the coupling-arc ends changes torque by less than 5.4×10^{-5} N m. These effects, and the material, tolerance and winding-area controls of Appendix F, are too small to explain the remaining cogging gap at the tested resolution.

E.1 Independent reference refinement

We rerun the public reference with its own degree-two isogeometric model, harmonic orders $3, 9, \ldots, 45$ and coupling torque, at 3, 6, 12 and 24 subdivisions in each patch direction. The 3×3 rerun first reproduces all 60 saved torques with differences at most 9.8×10^{-13} N m. The run uses GNU Octave 11.3.0 (Eaton et al., 2026), GeoPDEs (de Falco et al., 2011; Vázquez, 2016) at revision `53ae5ca1` and nurbs 1.4.3; compatibility edits, source hashes and raw records are retained. The corresponding field degrees of freedom are 4,079, 12,518, 43,220 and 159,920.

The reference cogging standard deviation changes from 0.01071 to 0.01220, 0.01212 and 0.01208 N m (Fig. 7a). GEODE L3 at order 45 gives 0.01209 N m, a 0.10% difference from the finest reference. The maximum pointwise discrepancy drops from 2.1×10^{-3} to 3.4×10^{-5} N m. At half and full current, the finest reference standard deviations are 0.03947 and 0.08014 N m; GEODE is 0.35% and 0.27% higher (panel b). All five finest-reference statistics, the three standard deviations and two loaded means, lie within GEODE's L3 refinement intervals. The coarse saved reference therefore accounts for the cogging mismatch in Section 3.4 at the matched interface order.

This agreement still concerns finite discretizations. The last cogging increments approximately halve; extending that trend would put a limit near 0.0120 N m, an estimate without a bound. The last two half-current standard-deviation increments are both about -0.3% and do not shrink; we therefore leave its remaining error unestimated. The 0.10% cogging agreement is a comparison of finite-resolution values, not an accuracy bound; neither refinement sequence is fully converged. Tightening the reference Newton stopping threshold from 10^{-6} to 10^{-14} changes the checked cogging torques by at most 4.4×10^{-8} N m. GEODE retains straight-sided geometry, and the high-field extrapolation of the iron law has not been varied in a paired torque experiment. Neither cross-code agreement nor refinement proves physical accuracy for the machine.

E.2 The saved optimized design

The saved reference design at iteration 120, denoted Opt120, includes 12 scalar parameters and ten grouped surface-control offsets. We evaluate its full exported geometry on freshly generated meshes; simply moving the finest initial mesh produces four inverted triangles. Its inner air-gap band no longer has the annular area assumed by the full-band Arkkio functional, so harmonic results here use coupling torque. Conforming results use only outer-band Arkkio on the fresh exported mesh. On the same L3, order-45 Opt120 field at full current, the outer-band and coupling estimates differ by about 1.3×10^{-4} N m in mean and 1.1×10^{-5} N m in standard deviation. This paired check supports the estimator substitution at this resolution. This estimator restriction is specific to these geometries; it does not validate outer-band Arkkio on arbitrary moved meshes or stator offsets.

With the evaluation current held fixed in the unscaled geometry, raising the transmitted order from 45 to 93 multiplies Opt120's F by 3.6, from 1.07×10^{-6} to 3.86×10^{-6} N^2 m^2 (Fig. 7c). The initial design's F rises by 16%. Its ratio to Opt120 is consequently 7,631 at order 45, 2,454 at order 93 and 2,444 for conforming coupling. The full-current Opt120 standard deviation rises from

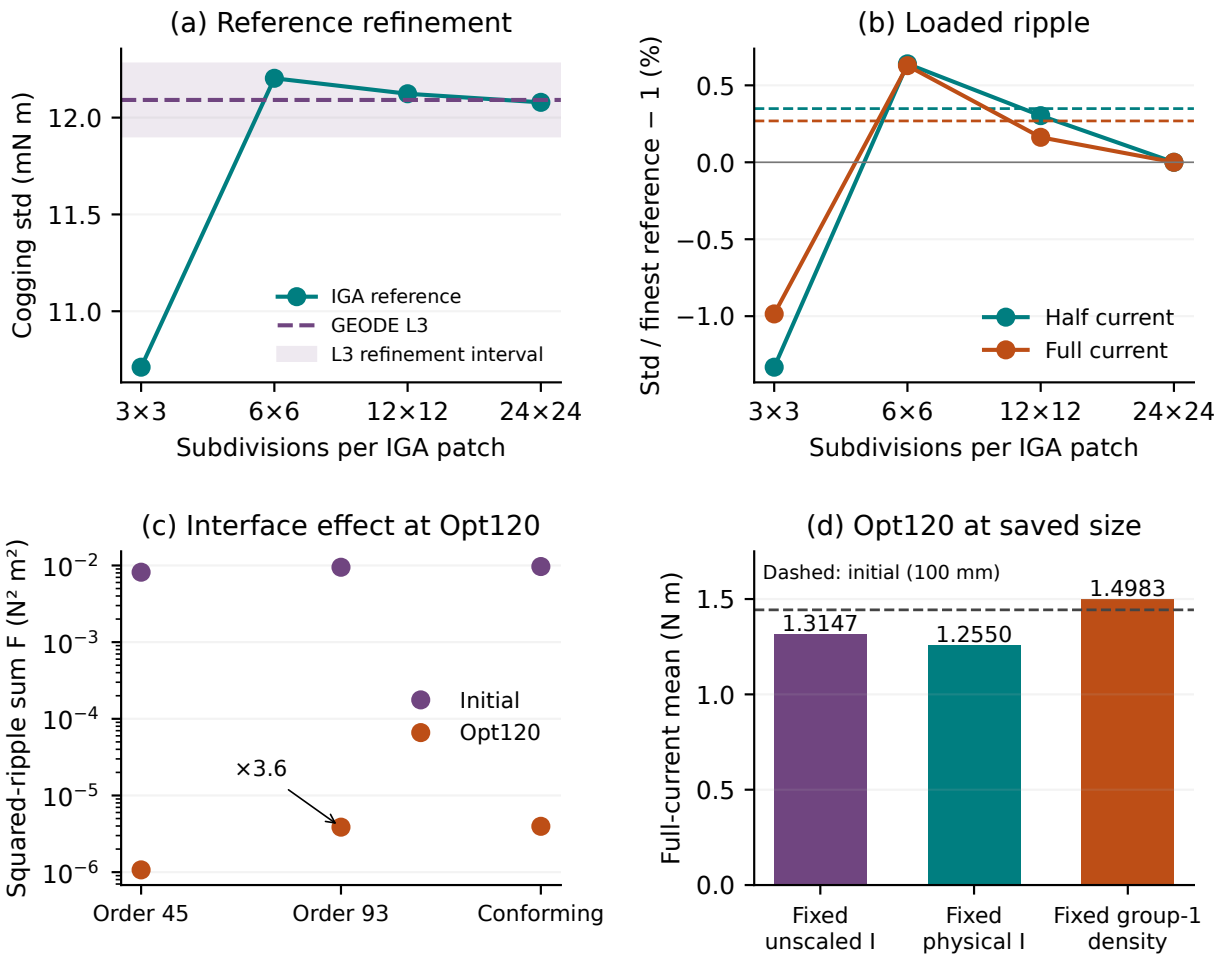


Figure 7: Reference and candidate validation from raw torque records. (a) Independent IGA reference cogging refinement at order 45; the dashed line is GEODE L3 and its shaded band uses Eq. (11). (b) Loaded standard deviation relative to the finest reference at each current; circles are IGA, horizontal lines are GEODE L3. The finest IGA point is zero by construction. Panels (a,b) use the initial geometry and 20 angles; GEODE uses Arkkio, with coupling torque checked separately. (c) Initial and Opt120 squared ripple sums at L3 on fresh meshes and 20 angles, under fixed unscaled current. Harmonic values use coupling torque; conforming values use outer-band Arkkio. (d) Full-current mean at order 45 for Opt120 under three current rules at its saved size (Appendix G); the dashed line is the initial mean at the initial size. All candidate values use coupling torque.

0.00066 to 0.00164 N m. The large ripple reduction survives these checks, but its magnitude depends strongly on the interface model: the order-93 change is about 96 times the last spatial-refinement increment of F (Appendix G).

A possible explanation is that optimization against order 45 favors cancellation within that truncated model. This is a conjecture, not an ablation result: increasing the order raises full-current ripple but lowers Opt120 cogging standard deviation from 0.00061 to 0.00035 N m. Appendix D reports the corresponding training-to-validation ratios at the designs selected in the optimization study. Opt120 is also outside the fixed-size, fixed-physical-current problem of Section 4: it shortens the stack from 100 to 80 mm and scales the radius by 1.0477, so its mean torque must be compared with the initial motor only after both size factors and the current rule are fixed. Appendix G gives

those normalizations, a reproduction comparison at the saved reference current, and the effect of angular sampling at this low ripple.

F Forward-model controls

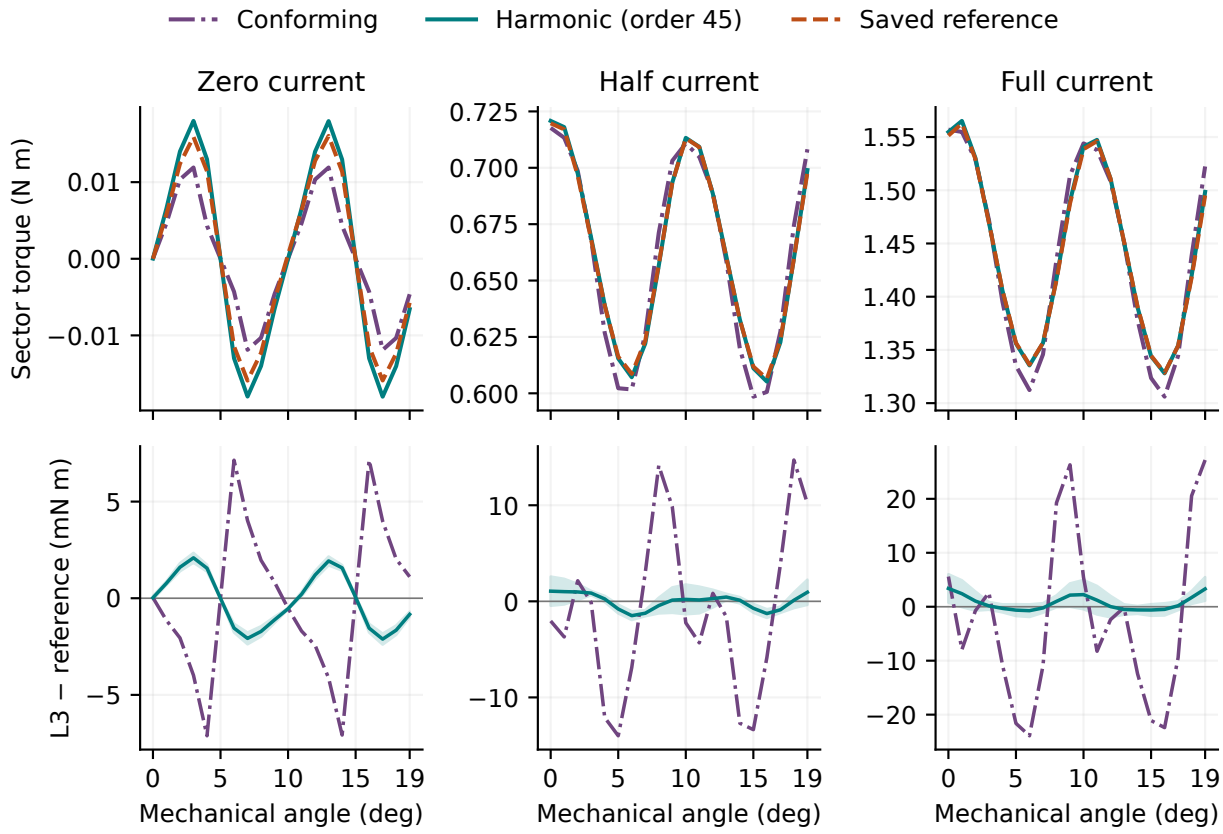


Figure 8: Top: L3 torque for conforming and order-45 harmonic coupling, compared with the saved reference. Bottom: each GEODE waveform minus the reference; shading is the harmonic model's pointwise estimated refinement interval from Eq. (11). Each residual panel has its own vertical scale, so compare residual magnitudes by their axis values, not by curve height. All GEODE curves use Arkkio torque, so the paired difference changes the coupling while holding the estimator fixed. The saved reference is coarse; Section E.1 reports its refinement.

The controlled changes in Table 11 are too small to explain the observed waveform gap at the tested settings. The near-zero material difference has particularly little effect on torque despite its 6.66% local reluctivity difference. This is a measured sensitivity of the initial design and does not establish insensitivity after optimization.

Table 11: Maximum absolute torque change relative to the baseline at the same mesh level, using conforming coupling and Arkkio torque. Entries summarize the recorded angle/current subsets.

Change	Levels	Max. $\lvert\Delta T\rvert$ (N m)
Reference low-field interpolation and clamp	L2, L3	2.1×10^{-6}
Relative Newton tolerance 10^{-12} vs 10^{-8}	L2; L3 full current	7.3×10^{-8}
Exact winding areas vs mesh areas	L2	3.9×10^{-4}
Inner or outer half of the Arkkio band	L3	7.3×10^{-5}

In the recorded estimator study, single-circle Maxwell-stress estimates vary more with the integration radius than the half-band Arkkio estimates. The paired coupling-torque and half-band checks of Section E avoid selecting a favorable radius. A post-processing filter of the solved conforming field also gives non-monotone ripple changes across the tested cuts and currents; it does not replace a field solve with a truncated interface. The complete filter results are retained in the accompanying forward-study records.

G Opt120: size, current policy and angular sampling

Between L2 and L3, changes in F, normalized by L3, reach 1.36% for the initial design and 2.71% for Opt120 across the three models on the exported mesh family; these increments are not error bounds. For Opt120 at order 45, the successive L1–L2 and L2–L3 increases in F are 3.05×10^{-8} and $2.90 \times 10^{-8}\,\mathrm{N^2\,m^2}$, so they barely shrink and do not establish an asymptotic rate. An alternate freshly constructed mesh family changes by 3.22% for Opt120 from L2 to L3. The $2.79 \times 10^{-6}\,\mathrm{N^2\,m^2}$ interface effect is about 96 times the last export-mesh increment, despite this unresolved remainder.

Current policy changes Opt120's feasibility (Fig. 7d). At L3 and order 45, the initial full-current mean is 1.4433 N m. Opt120 gives 1.3147 N m under fixed *unscaled* current, 1.2550 N m under fixed physical current, and 1.4983 N m under the reference's fixed group-1 density rule. The first two are 8.9% and 13.0% below the initial mean; the density-rule value is 3.8% above it. These policy comparisons use the same Opt120 geometry, but comparisons with the initial motor also change size: Opt120 shortens the stack from 100 to 80 mm and has radial scale $s = 1.0477$, unscaled outer diameter 130.293 mm and operating angle $-11.646°$ (Table 13). Every Opt120 torque carries the explicit size factor $s^2 L/(0.1\,\mathrm{m}) = 0.878175$. Removing both size factors from the fixed-unscaled-current mean gives 1.4971 N m, 3.7% above the initial mean. Applying the same normalization to fixed physical current gives 1.4291 N m, 1.0% below the initial mean; the density-rule value becomes 1.7061 N m, 18.2% above it. Normalizing only length to 100 mm under fixed unscaled current gives 1.6434 N m (+13.9%), retaining the radial factor $s^2 = 1.098$. These algebraic normalizations separate the size factors from the current rule; they are not reoptimized designs. Most of the saved-size deficit therefore comes from the size factor, although the normalized fixed-physical-current value remains lower.

A separate reproduction comparison uses the saved reference current at each design, rather than fixed unscaled current. At Opt120, GEODE's order-45 full-current mean is 1.4986 N m versus 1.5000 N m in the saved arrays, but its standard deviation is 3.53×10^{-4} versus 1.76×10^{-4} N m, a factor of 2.0. The half-current values are 5.53×10^{-4} and 4.37×10^{-4} N m. The largest pointwise difference across the two loaded currents is about 2.1×10^{-3} N m. With this matched current policy and zero-current values included (zero current is rule-independent and taken from the fixed-current record), the initial-to-Opt120 objective ratios are 10,170 for GEODE and 11,982 for the saved reference. The 7,631 ratio of Section E.2 instead holds unscaled current fixed. Reference refinement was run only at the initial design, so the cause of the Opt120 discrepancy remains unresolved.

Angular sampling becomes more consequential at this low ripple. Relative to the 40-angle grid, the ten-angle Opt120 objective reads about 1.05% high at order 45 and 5.90% low at order 93 on the fresh L3 mesh; the conforming value reads 17.1% low. The accompanying records retain shifted 20-angle grids and withheld currents 0.25 and 0.75. Candidate validation must check these effects at the candidate, since initial-design agreement alone would miss them. All 3,240 solve rows in the 18 successful records used for these candidate comparisons stopped at tolerance. Figure 9 shows the full and mean-subtracted waveforms.

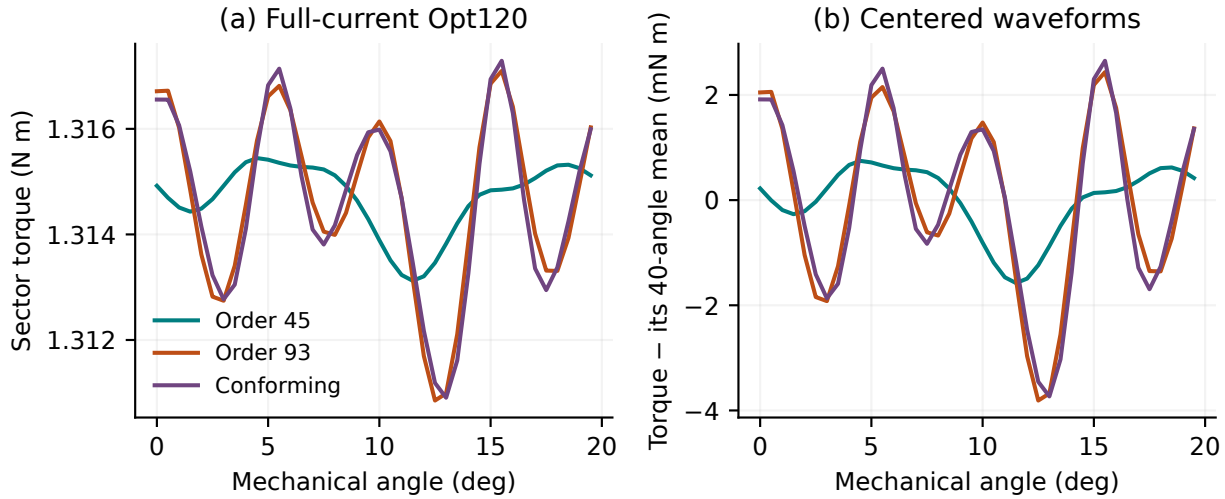


Figure 9: Opt120 full-current torque on fresh L3 meshes at 40 angles, with fixed unscaled current. Harmonic orders 45 and 93 use coupling torque; conforming coupling uses outer-band Arkkio. Left: full torque, retaining each model's mean. Right: each curve minus its own 40-angle mean. The interface change mainly alters the resolved ripple.

H Gradient verification details and derivative cost

Excitation rules

In the scalar derivative study, fixed current means fixed peak current in the unscaled geometry, so $J_g = 12 s_g I_q / |\Omega_g(X)|$ changes when a slot changes area. The reference optimizer's rule, denoted *fixed group-1 density*, sets $I_* \propto |\Omega_1(X)| s$, where s is radial scale, with the proportionality constant fixed at the initial design; the unscaled geometry is evaluated and torque is multiplied by s^2, following the reference. Both rules agree at the initial geometry but have different derivatives. Group-2 density is not held fixed because the ratio of the two winding areas can change. Rotor parameters MAG, MH and MW leave winding area unchanged and therefore give identical derivatives under the two rules. The fixed-size problem sets radial scale to one, making fixed physical and unscaled current coincide.

Taylor remainders

After removing duplicates and the analytic axial-length and radial-scale cases, 530 of 592 loaded comparisons have an anchored quadratic-remainder window spanning at least four decades in step size; 388 span at least five. The window is the contiguous run where remainder$/h^2$ stays within a factor of two of its value at $h = 10^{-5}$ m (or 10^{-2} degrees for operating angle). Smaller gradients reach round-off sooner. An anchored-window diagnostic fails for the initial-design, full-current conforming/fixed-density SW1 ripple cases at L0 and L1: at L0 the remainder changes sign near the anchor, whereas at L1 remainder$/h^2$ is still drifting there. Their widest quadratic windows span two

decades. The nearly cancelling zero-current mean is a weaker check: its best relative errors reach 7.0×10^{-4}. Cogging standard-deviation errors are at most 8.9×10^{-7}. Thus the loaded agreement does not describe every zero-current derivative. Discrete derivative accuracy also does not imply mesh convergence, as the next paragraph shows.

Model dependence of verified sensitivities

The two model–estimator pairs give different ripple sensitivities even at the same design (Fig. 10). At L2 and full fixed current, the MAG ripple derivative is 1.28 N m/m for harmonic/coupling torque versus 1.94 for conforming/Arkkio at the initial design. At the perturbed design the values are 0.75 and 1.87 N m/m. These differences persist under mesh refinement. At L3 the MAG pairs are 1.22 versus 1.87 at the initial design and 0.57 versus 1.67 N m/m at the perturbed design. In particular, the latter harmonic component changes appreciably even though the box-scaled vector changes by only 1.19% of its L2 norm; the norm itself decreases by 0.19%. Mean-torque sensitivities agree more closely in the box-scaled comparison: the Euclidean norm of the vector difference, divided by the conforming vector norm, is 0.58% and 0.66% for the mean at the initial and perturbed designs, versus 6.7% and 10.2% for ripple. The norm weights the largest components most heavily: the initial SW2 and SW3 mean components differ by 6.5% and 16.5% relative to conforming values. At L3, the ripple gaps remain 6.58% and 10.25%, while each model changes by 0.46–1.19% from L2 to L3. The mean gaps fall to 0.16% and 0.26%; the conforming model itself changes by 0.52–0.54%. The L2 mean gap is therefore comparable to spatial refinement, whereas the ripple gap persists beyond the measured refinement increments. Because the estimator changes with the model, this experiment measures their combined effect; it does not isolate an interface-only derivative error or establish which gradient is closer to the physical machine.

Discrete derivative accuracy also does not imply mesh convergence. In the harmonic model, SW3 mean-torque sensitivities change sign between coarse levels and their increments shrink with refinement; ripple gradients converge more slowly than most mean gradients. The conforming/Arkkio SW2 mean sensitivity is non-monotone across L0–L3 (-46.5, -41.3, -44.4, -47.2 N m/m), whereas harmonic/coupling torque approaches -47.1 from -49.8 N m/m. Fresh-remesh checks of MW, SW1 and SW2 are limited at small steps by changes in connectivity. The L2 MW mean-derivative fit has relative gap 2.6×10^{-6}, but individual remeshed steps at $h \geq 10^{-4}$ m scatter by about 6×10^{-4}. The largest L2 full-current remeshed-versus-moved fitted discrepancy is 3.8% for the SW1 ripple derivative. These checks support the discrete chain rule while leaving refinement of candidate designs necessary.

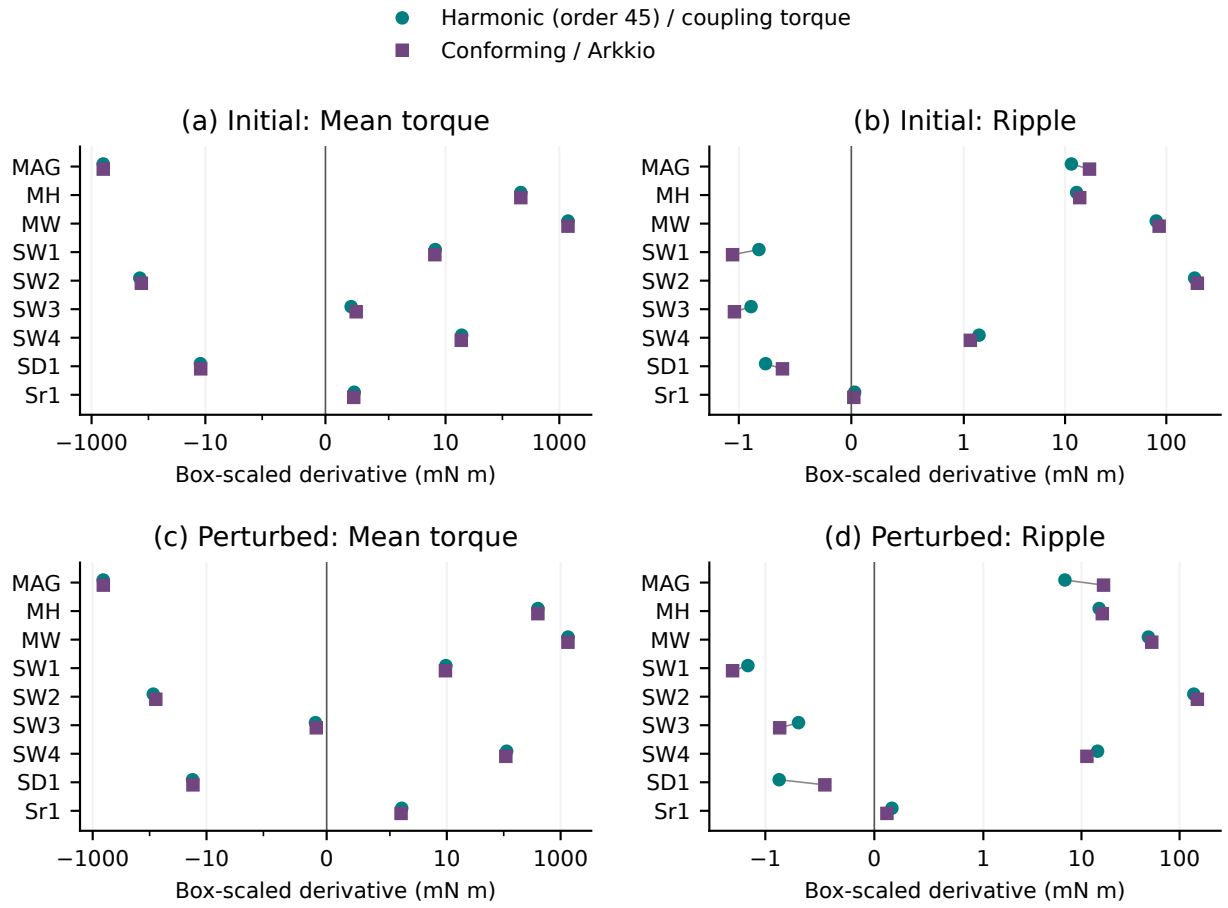


Figure 10: Full fixed-current sensitivities at L2, scaled by each geometric parameter's reference box width w_k. Top: initial design; bottom: perturbed design. L3 MAG values and refinement changes are given in the text. Left: mean torque; right: standard deviation. The plotted quantity $w_k \partial q / \partial p_k$ is a local sensitivity, not a finite change across the box. Coupling and torque estimator both differ between the two series. Gray segments join the two estimates for each parameter. Horizontal axes are symmetric-log, linear within ± 1 mN m. All nine widths, in mm, are 9, 10, 15, 4, 3, 1, 18, 70 and 1.5 in the displayed order.

The recorded timings use an Apple M3 Ultra with 28 cores, default solver threading, and concurrent host load averaging 14–24. The effective thread count was not pinned. A forward evaluation covers ten angles, starts from zero, and warm-starts successive angles. The full gradient includes that forward, ten torque adjoints, mesh derivatives and contractions for all 12 parameters and both aggregate objectives. Central differences use 24 forward sweeps, each warm-started from the base states, at step $h = 10^{-6}$ in each parameter's stored unit (m, degrees or dimensionless scale), plus the shared base forward for a common starting point. Across L0–L2, both couplings and tolerances 10^{-8} and 10^{-14}, the full gradient costs 1.13–1.20 forwards, whereas central differences cost 8.2–10.5 including the shared base solve (Fig. 6d and Table 12). Shape-map setup is a separate one-time cost of these timings; the optimization study charges whole-run wall time, so it is included there. These derivative timings concern one loaded host and are a separate workload from the 1,640 forward validation solves of Section E.

Table 12: Recorded harmonic-model cost at relative tolerance 10^{-14}, initial design and full current. Times are seconds for ten-angle sweeps; FD includes the shared base sweep, and setup is a one-time shape-map cost.

Mesh	Forward	Forward + gradient	Base + FD	Setup
L0	0.664	0.748	6.997	0.011
L1	3.174	3.658	32.811	0.041
L2	12.702	14.644	129.390	0.222

I Linear slotless screening control

The stored slotless model has 24 distinct angles. Fitting $T(\phi) = c_0 + c_1 \cos\phi + c_2 \sin\phi$, with electrical angle $\phi = p\theta_r$, mechanical rotor angle θ_r and pole-pair count p, on alternate angles and evaluating on the other twelve gives normalized held-out RMSE 3.08×10^{-10} and maximum error 6.82×10^{-10}, using the recorded analytic-amplitude normalization. This is a locked-rotor angle sweep, not synchronous operating ripple. For a fixed linear field operator and sinusoidal sources, the field lies in a small basis of source responses; a quadratic torque functional can contain only the corresponding finite set of harmonics. The particular model needs just the three fitted coefficients. This is an angular interpolation test at one geometry, not a shape-gradient test or evidence of generalization across designs.

J Design points for derivative checks

Table 13 identifies the scalar derivative-study coordinates and the saved Opt120 coordinates. Values are rounded for display; the accompanying records retain full precision. The operating angle is stored in degrees and converted to radians in the current equation. These coordinates describe the scalar design space; the saved reference optimum additionally contains surface-control offsets.

Table 13: Initial and interior perturbed design points for derivative checks, alongside Opt120 for the forward validation. Lengths refer to the unscaled geometry; surface-control offsets are retained in the data.

Parameter	Meaning	Unit	Initial	Perturbed	Opt120
MAG	Pole-axis iron depth above magnet	mm	7.0000	6.5506	6.1011
MH	Radial magnet height	mm	7.0000	4.8722	2.7444
MW	Magnet width	mm	19.0000	19.8670	20.7340
SW1	Stator tooth width	mm	4.0000	4.0309	4.0619
SW2	Slot-opening width	mm	2.3000	1.6500	1.0001
SW3	Slot-opening depth	mm	1.0000	0.7502	0.5004
SW4	Back-iron depth on slot axis	mm	8.2500	6.8418	5.4336
SD1	Stator outer diameter	mm	135.0000	132.6466	130.2931
Sr1	Slot-bottom fillet radius	mm	1.0000	1.4999	1.9998
LENGTH	Axial length	mm	100.0000	90.0003	80.0005
ϕ_0	Electrical current offset	°	0.0000	−5.8231	−11.6463
ScaleR	Radial similarity scale	1	1.0000	1.0239	1.0477

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